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LC/MS analysis of Plasma and CSF samples from PPMI

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PPMI project: 180

Version 3 – August 2024

Summary

A comprehensive profiling of metabolites and lipids in plasma and CSF samples from PPMI was conducted to uncover metabolic phenotypes intricately connected to the genetics and pathology of Parkinson's disease. The analysis included sphingolipids and polyamines, key indicators of disrupted lysosomal functions, along with markers of caffeine and purine metabolism, and other indicators of inflammatory and oxidative stress.

Revision 1. Cross-contribution Compensating Multiple Standard Normalization (CCMN) normalization removed:

In version 1, endogenous analyte areas were first batch-corrected, followed by further adjustment using the CCMN method as described in Reference 1. However, post-blinding analysis revealed aberrant bimodal distributions across all subgroups, which were not present when the IS normalization factor was removed. Based on this observation, the decision was made to eliminate the CCMN normalization and replace it with an area ratio approach. In this new approach, endogenous analyte areas were normalized to specific surrogate spiked-in internal standards. Tables 1 and 2 provide the specific pairings of endogenous analytes to surrogate internal standards used for generating area ratios.

Revision 2. Replaced batch-adjusted area with unadjusted raw area:

In revision 2, analyte abundance data were presented as unadjusted area (UNIT = area), unadjusted area ratio (UNIT = area_ratio), and batch-adjusted area ratio (UNIT = adjusted_area_ratio). Batch adjustments were performed using the method described in Reference 2

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

1) Redestig, et.al., Anal Chem, 2009, 81, 7974-7980
2) Leek JT et. Al., Bioinformatics, 2012, 28, 882 - 883

Revision 3. Added method description for CSF samples and revised plasma method description to improve clarity.

Plasma Testing Design

Both targeted metabolomic/lipidomic and untargeted metabolomic analyses were conducted across eight different batches. Samples in each batch were selected to ensure that independent factors and covariates of interest were evenly distributed among the batches and randomized. Additionally, PPMI created three pooled control and three pooled PD case plasma samples, which were evenly distributed within and between batches. Each pooled sample was injected 3-6 times per batch, and values from replicate extractions and injections were used to calculate assay CV and inter-batch deviations.

Each extracted sample was sequentially analyzed using the following methods: metabolomics (positive and negative ionization modes), lipidomics (positive and negative ionization modes), GlcCer/GalCer panel (positive ionization), GM panel (negative ionization), BMP panel (negative ionization), and untargeted metabolomics (positive and negative ionization modes). Detailed descriptions of the sample extraction, analysis, and data reporting procedures are provided below.

Methods

Plasma sample preparation for LC/MS analysis

Thawed plasma samples were centrifuged at 3.5k x g for 10 minutes at 4°C. A 10 µL aliquot of plasma was transferred to an Agilent Captiva collection plate and processed using the Agilent Bravo Metabolomics Sample Prep Platform (Agilent Technologies, Santa Clara, CA), following the Bravo Low Volume Plasma Metabolite/Lipid protocols. A 1:1 methanol/ethanol (v/v) solution containing internal standards (112.5 µL) was added to each sample. The plates were shaken for 1 minute at 1000 rpm and incubated at room temperature for 10 minutes. Next, 82.5 µL of water was added to each well, followed by shaking for 1 minute at 500 rpm and a 10-minute incubation at room temperature. The samples (200 µL) were then transferred to a Captiva EMR-Lipid plate to collect the metabolite filtrate and remove proteins and lipids. The Captiva EMR-Lipid plate was washed twice with 250 µL of a 1:1:1 water/ethanol/methanol (v/v/v) solution. The metabolite filtrate was dried under N2 gas for 4 hours and reconstituted with 200 µL of methanol.

To collect the lipids retained on the Captiva EMR-Lipid plate, 1.8 mL of a 1:2 dichloromethane/methanol (v/v) solution was added to the plate for elution. The lipid filtrate was collected in a 2 mL glass-coated microplate, dried under N2 gas for 4 hours, and reconstituted with 200 µL of methanol. For the analysis of the GlcCer/GalCer panel, a 25 µL aliquot from the methanol lipid fraction was dried under N2 gas for 4 hours and resuspended in 50 µL of a 92.5/5/2.5 acetonitrile/isopropanol/water solution containing 5 mM ammonium formate and 0.5% formic acid.

LCMS targeted analysis of lipids in plasma

Lipid analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+). For each analysis, 5 µL of sample was injected on a BEH C18 1.7 µm, 2.1×100 mm column (Waters) using a flow rate of 0.25 mL/min at 55°C. For positive ionization mode, mobile phase A consisted of 60:40 acetonitrile/water (v/v) with 10 mM ammonium formate + 0.1% formic acid; mobile phase B consisted of 90:10 isopropyl alcohol/acetonitrile (v/v) with 10 mM ammonium formate + 0.1% formic acid. For negative ionization mode, mobile phase A consisted of 60:40 acetonitrile/water (v/v) with 10 mM ammonium acetate + 0.1% acetic acid; mobile phase B consisted of 90:10 isopropyl alcohol/acetonitrile (v/v) with 10 mM ammonium acetate + 0.1% acetic acid. The gradient was programmed as follows: 0.0-8.0 min from 45% B to 99% B, 8.0-9.0 min at 99% B, 9.0-9.1 min to 45% B, and 9.1-10.0 min at 45% B. Electrospray ionization was performed in positive or negative ion mode. For the QTRAP 6500+, we applied the following settings: curtain gas at 30 psi (negative mode) and curtain gas at 40 psi (positive mode); collision gas was set at medium; ion spray voltage at 5500 V (positive mode) or -4500 V (negative mode); temperature at 250°C (positive mode) or 600°C (negative mode); ion source Gas 1 at 55 psi; ion source Gas 2 at 60 psi; entrance potential at 10 V (positive mode) or -10 V (negative mode); and collision cell exit potential at 12.5 V (positive mode) or -15.0 V (negative mode). Data acquisition was performed in multiple reaction monitoring mode (MRM) with the collision energy (CE) values reported in Tables 1 and 2. Lipids were quantified using a mixture of non-endogenous internal standards as reported in Table 1 and 2. Quantification was performed using MultiQuant 3.02 (Sciex).

PARKINSON'S PROGRESSION MARKERS INITIATIVE

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Table 1: Lipidomics in positive mode parameters

Name	Internal Standard	Q1 m/z	Q3 m/z	CE (V)	% Samples with missing area ratio (Plasma)
1-O-Palmitoyl-Cer(d18:1/18:0)	Cer(d18:1/16:0(d7))	786.8	502.5	35	100
24-Hydroxycholesterol	24-Hydroxcholesterol-d7	385.3	367.3	30	100
24-Hydroxycholesterol(d7)	IS	392.3	367.3	30	NA
3-O-SulfoLacCer(d18:1/18:0)	LacCer(d18:1/17:0)	970.8	548.5	61	100
4-beta-Hydroxycholesterol	Cholesterol(d7)	420.3	385.3	15	55.3
7-keto-Cholesterol	Cholesterol(d7)	401.3	383.3	15	0.2
CE HETE	CE(18:1(d7))	706.6	369.2	25	100
CE HODE	CE(18:1(d7))	682.6	369.2	25	100
CE HpODE	CE(18:1(d7))	698.6	369.2	25	100
CE oxoHETE	CE(18:1(d7))	704.6	369.2	25	100
CE oxoODE	CE(18:1(d7))	680.6	369.2	25	100
CE(16:1)	CE(18:1(d7))	640.6	369.3	26	0
CE(18:1(d7))	IS	675.2	369.4	26	NA
CE(18:1)	CE(18:1(d7))	668.6	369.3	26	0
CE(18:2)	CE(18:1(d7))	666.6	369.3	26	0
CE(20:4)	CE(18:1(d7))	690.6	369.3	26	0
CE(20:5)	CE(18:1(d7))	688.6	369.3	26	0
CE(22:6)	CE(18:1(d7))	714.6	369.3	26	0
Cer(d18:0/16:0)	Cer(d18:1/16:0(d7))	540.6	284.3	40	100
Cer(d18:0/18:0)	Cer(d18:1/16:0(d7))	568.7	284.3	40	100
Cer(d18:0/24:0)	Cer(d18:1/16:0(d7))	652.9	284.3	40	100
Cer(d18:0/24:1)	Cer(d18:1/16:0(d7))	650.9	284.4	40	100
Cer(d18:1/16:0(d7))	IS	545.5	271.4	40	NA
Cer(d18:1/16:0)	Cer(d18:1/16:0(d7))	538.5	264.3	40	0
Cer(d18:1/18:0)	Cer(d18:1/16:0(d7))	566.6	264.3	40	0
Cer(d18:1/24:0)	Cer(d18:1/16:0(d7))	650.6	264.3	40	0
Cer(d18:1/24:1)	Cer(d18:1/16:0(d7))	648.6	264.3	40	0

PARKINSON'S PROGRESSION MARKERS INITIATIVE

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Cholesterol	Cholesterol(d7)	369.3	369.3	10	0.1
Cholesterol(d7)	IS	376.2	376.2	10	NA
Cholesteryl hexoside	CE(18:1(d7))	566.6	369.3	17	0
Coenzyme Q10	TG(15:0/18:1(d7)/15:0)	863.3	197.2	35	0
DG(15:0/18:1(d7))	IS	605.6	346.5	30	NA
DG(16:0_18:1)	DG(15:0/18:1(d7))	612.4	313.3	30	0
DG(16:0_20:4)	DG(15:0/18:1(d7))	634.5	313.3	30	0
DG(18:0_18:1)	DG(15:0/18:1(d7))	640.4	341.3	30	0
DG(18:0_20:4)	DG(15:0/18:1(d7))	662.5	341.3	30	0
DG(18:0_22:6)	DG(15:0/18:1(d7))	686.6	341.3	30	0.1
DG(18:1_20:4)	DG(15:0/18:1(d7))	660.5	339.3	30	0
DG(18:1/18:1)	DG(15:0/18:1(d7))	638.4	339.3	30	0
GB3(d18:1/16:0)	GB3(d18:1/18:0(d3))	1025	520.5	40	0
GB3(d18:1/18:0(d3))	IS	1056	551.6	40	NA
GB3(d18:1/18:0)	GB3(d18:1/18:0(d3))	1053	548.6	40	0
GB3(d18:1/24:0)	GB3(d18:1/18:0(d3))	1137	632.6	40	0
GB3(d18:1/24:1)	GB3(d18:1/18:0(d3))	1135	630.6	40	0
GlcCer(d18:1(d5)/18:0)	IS	733.6	269.3	45	NA
GlcCer(d18:1/16:0(d3))	IS	703.7	264.3	51	NA
Glucosylsphingosine(d5)	IS	467.2	269.3	16	NA
HexCer(d18:1/12:0)	GlcCer(d18:1(d5)/18:0)	644.5	264.3	40	100
HexCer(d18:1/16:0)	GlcCer(d18:1(d5)/18:0)	700.6	264.3	40	0
HexCer(d18:1/18:0)	GlcCer(d18:1(d5)/18:0)	728.6	264.3	40	0
HexCer(d18:1/22:0)	GlcCer(d18:1(d5)/18:0)	784.7	264.4	40	0
HexCer(d18:1/24:0)	GlcCer(d18:1(d5)/18:0)	812.7	264.3	40	0
HexCer(d18:1/24:1)	GlcCer(d18:1(d5)/18:0)	810.7	264.3	40	0
Hexosylsphingosine	Glucosylsphingosine(d5)	462.3	264.2	16	100
LacCer(d18:1/16:0)	LacCer(d18:1/17:0)	862.6	264.3	40	0
LacCer(d18:1/17:0)	IS	876.6	264.3	40	NA
LacCer(d18:1/18:0)	LacCer(d18:1/17:0)	890.7	264.3	40	0
LacCer(d18:1/24:0)	LacCer(d18:1/17:0)	974.8	264.3	40	0
LacCer(d18:1/24:1)	LacCer(d18:1/17:0)	972.7	264.3	40	0

Lactosylsphingosine	Glucosylsphingosine(d5)	624.4	264.3	16	100
LPC(16:0)	LPC(18:1(d7))	496.3	184.1	40	0
LPC(16:1)	LPC(18:1(d7))	494.5	184.1	40	0
LPC(18:0)	LPC(18:1(d7))	524.3	184.1	40	0
LPC(18:1(d7))	IS	529.3	184.1	40	NA
LPC(18:1)	LPC(18:1(d7))	522.3	184.1	40	0
LPC(20:4)	LPC(18:1(d7))	544.3	184.1	40	0
LPC(22:6)	LPC(18:1(d7))	568.3	184.1	40	0
LPC(24:0)	LPC(18:1(d7))	608.5	184.1	40	0
LPC(24:1)	LPC(18:1(d7))	606.5	184.1	40	0
LPC(26:0)	LPC(18:1(d7))	636.5	104.1	40	0.1
LPC(26:1)	LPC(18:1(d7))	634.5	104.1	40	24.8
lyso-GB3	lyso-GB3-d7	786.6	264.3	46	100
lyso-GB3-d7	IS	793.5	271.3	46	NA
lyso-GB4	GB3(d18:1/18:0(d3))	990.6	264.3	52	100
MG(16:0)	MG(18:1(d7))	348.3	239.3	22	24.8
MG(16:1)	MG(18:1(d7))	346.3	237.3	22	100
MG(18:0)	MG(18:1(d7))	376.3	267.3	22	12.6
MG(18:1(d7))	IS	381.3	272.5	22	NA
MG(18:1)	MG(18:1(d7))	374.3	265.3	22	0
MG(20:4)	MG(18:1(d7))	396.3	287.3	22	100
N-Oleoylethanolamine	LPC(18:1(d7))	326.3	62.1	23	2
N-Palmitoyl-O-phosphocholineserine	LPC(18:1(d7))	509.5	184.1	40	100
Palmitoylethanolamine	LPC(18:1(d7))	300.3	62.1	23	0.2
PC(15:0/18:1(d7))	IS	754.6	184.1	40	NA
PC(16:0/5:0(CHO))	PC(15:0/18:1(d7))	594.5	184.1	40	0
PC(16:0/9:0(CHO))	PC(15:0/18:1(d7))	650.4	184.1	40	0
PC(16:0/9:0(COOH))	PC(15:0/18:1(d7))	666.4	184.1	40	0
PC(18:0/20:4(OH[S]))	PC(15:0/18:1(d7))	826.6	184.1	40	100
PC(18:0/20:4(OOH[S]))	PC(15:0/18:1(d7))	842.6	184.1	40	100
PC(34:1)	PC(15:0/18:1(d7))	760.6	184.1	40	0
PC(36:1)	PC(15:0/18:1(d7))	788.6	184.1	40	0

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PC(36:2)	PC(15:0/18:1(d7))	786.6	184.1	40	0
PC(36:4)	PC(15:0/18:1(d7))	782.6	184.1	40	0
PC(38:4)	PC(15:0/18:1(d7))	810.6	184.1	40	0
PC(38:6)	PC(15:0/18:1(d7))	806.6	184.1	40	0
PC(40:5)	PC(15:0/18:1(d7))	836.6	184.1	40	0
PC(40:6)	PC(15:0/18:1(d7))	834.6	184.1	40	0
PC(O-16:0/0:0)	LPC(18:1(d7))	482.3	184.1	40	0
PC(O-16:0/2:0)	LPC(18:1(d7))	524.3	184.2	40	0
PC(O-18:0/2:0)	LPC(18:1(d7))	552.5	184.1	40	0
PE(15:0/18:1(d7))	IS	711.6	570.5	40	NA
PE(18:0/20:4(OH[S]))	PE(15:0/18:1(d7))	784.5	643.4	40	100
PE(18:0/20:4(OOH[S]))	PE(15:0/18:1(d7))	800.5	659.4	40	100
PE(34:1)	PE(15:0/18:1(d7))	718.6	577.6	40	0
PE(36:1)	PE(15:0/18:1(d7))	746.6	605.5	40	0
PE(36:2)	PE(15:0/18:1(d7))	744.6	603.5	40	0
PE(36:4)	PE(15:0/18:1(d7))	740.6	599.5	40	0
PE(38:4)	PE(15:0/18:1(d7))	768.6	627.5	40	0
PE(38:5)	PE(15:0/18:1(d7))	766.6	625.5	40	0
PE(38:6)	PE(15:0/18:1(d7))	764.6	623.5	40	0
PE(40:4)	PE(15:0/18:1(d7))	796.6	655.5	40	0
PE(40:5)	PE(15:0/18:1(d7))	794.6	635.5	40	100
PE(40:6)	PE(15:0/18:1(d7))	792.6	651.5	40	0
PE(40:7)	PE(15:0/18:1(d7))	790.6	649.5	40	0
Sitosteryl hexoside	CE(18:1(d7))	594.6	397.4	17	0.1
SM(d18:1(d9)/18:1)	IS	738.7	184.1	40	NA
SM(d18:1/16:0)	SM(d18:1(d9)/18:1)	703.6	184.1	40	0
SM(d18:1/18:0)	SM(d18:1(d9)/18:1)	731.6	184.1	40	0
SM(d18:1/24:0)	SM(d18:1(d9)/18:1)	815.7	184.1	40	0
SM(d18:1/24:1)	SM(d18:1(d9)/18:1)	813.7	184.1	40	0
Sphinganine	Sphingosine(d17:1)	302.2	284.3	20	2.9
Sphinganine 1-phosphate	Sphingosine 1-phosphate-d7	382.3	284.3	18	100
Sphingosine	Sphingosine(d17:1)	300.2	264.3	20	100

Sphingosine 1-phosphate	Sphingosine 1-phosphate-d7	380.3	264.3	25	100
Sphingosine 1-phosphate-d7	IS	387.3	271.3	25	NA
Sphingosine 1-phosphocholine	LPC(18:1(d7))	465.5	184.1	40	100
Sphingosine(d17:1)	IS	286.2	250.3	20	NA
TG(15:0/18:1(d7)/15:0)	IS	829.8	570.8	40	NA
TG(18:0_36:2)	TG(15:0/18:1(d7)/15:0)	904.7	603.4	40	0
TG(18:1_34:2)	TG(15:0/18:1(d7)/15:0)	874.7	575.4	40	0
TG(18:1_34:3)	TG(15:0/18:1(d7)/15:0)	872.7	573.4	40	0
TG(20:4_32:1)	TG(15:0/18:1(d7)/15:0)	870.6	549.3	40	0
TG(20:4_34:2)	TG(15:0/18:1(d7)/15:0)	896.6	575.3	40	0
TG(20:4_34:3)	TG(15:0/18:1(d7)/15:0)	894.6	573.3	40	0
TG(20:4_36:0)	TG(15:0/18:1(d7)/15:0)	928.8	607.5	40	0.1
TG(20:4_36:2)	TG(15:0/18:1(d7)/15:0)	924.7	603.4	40	0
TG(20:4_36:3)	TG(15:0/18:1(d7)/15:0)	922.7	601.4	40	0
TG(22:6_36:2)	TG(15:0/18:1(d7)/15:0)	948.7	603.4	40	0
TG(22:6_38:1)	TG(15:0/18:1(d7)/15:0)	978.7	633.4	40	0.8
TG(22:6_38:2)	TG(15:0/18:1(d7)/15:0)	976.7	631.4	40	0.1

Note: IS - Internal standard;

The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples.

Table 2: Lipidomics in negative mode parameters

Name	**Internal Standard**	**Q1 m/z**	**Q3 m/z**	**CE (V)**	**% Samples with missing area ratio (Plasma)**
(3-O-sulfo)GalCer(d18:1/16:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	778.5	97	-150	0
(3-O-sulfo)GalCer(d18:1/18:0(2OH)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	822.6	97	-150	0
(3-O-sulfo)GalCer(d18:1/18:0(d3))	IS	809.6	97	-150	NA
(3-O-sulfo)GalCer(d18:1/18:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	806.6	97	-150	0
(3-O-sulfo)GalCer(d18:1/24:0(2OH))	(3-O-sulfo)GalCer(d18:1/18:0(d3))	906.7	97	-150	0
(3-O-sulfo)GalCer(d18:1/24:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	890.7	97	-150	0
(3-O-sulfo)GalCer(d18:1/24:1(2OH))	(3-O-sulfo)GalCer(d18:1/18:0(d3))	904.7	97	-150	0
(3-O-sulfo)GalCer(d18:1/24:1)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	888.7	97	-150	0
Arachidonic acid	Arachidonic acid-d8	303.2	303.2	-10	100

PARKINSON'S PROGRESSION MARKERS INITIATIVE

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Arachidonic acid_MRM	Arachidonic acid-d8_MRM	303.2	259.1	-19	0.1
Arachidonic acid-d8	IS	311.3	311.3	-10	NA
Arachidonic acid-d8_MRM	IS	311.3	267.1	-19	NA
BMP(14:0/14:0)	IS	665.3	227.2	-50	NA
BMP(16:0_18:1)	BMP(14:0/14:0)	747.5	255.4	-50	0.2
BMP(16:0_20:4)	BMP(14:0/14:0)	769.5	255.4	-50	100
BMP(16:0_22:6)	BMP(14:0/14:0)	795.5	255.4	-50	100
BMP(16:1/16:1)	BMP(14:0/14:0)	717.5	253.1	-50	100
BMP(18:0_18:1)	BMP(14:0/14:0)	775.5	281.4	-50	0.1
BMP(18:0_20:4)	BMP(14:0/14:0)	797.5	283.4	-50	100
BMP(18:0_22:6)	BMP(14:0/14:0)	823.5	283.4	-50	100
BMP(18:1/18:1)	BMP(14:0/14:0)	773.5	281.3	-50	0
BMP(20:4/20:4)	BMP(14:0/14:0)	817.5	303.3	-50	100
BMP(22:6/22:6)	BMP(14:0/14:0)	865.5	327.3	-50	6.5
Cholesterol sulfate	(3-O-sulfo)GalCer(d18:1/18:0(d3))	465.3	96.7	-80	0
CL(14:0/14:0/14:0/14:0)	IS	619.5	227.2	-50	NA
CL(72:6/18:2)	CL(14:0/14:0/14:0/14:0)	725.7	279.2	-50	100
CL(72:7/18:2)	CL(14:0/14:0/14:0/14:0)	724.7	279.2	-50	100
CL(72:8-2(OOH)/18:2)	CL(14:0/14:0/14:0/14:0)	755.7	279.2	-50	0
CL(72:8/18:2)	CL(14:0/14:0/14:0/14:0)	723.7	279.3	-50	100
CL(74:9/18:2)	CL(14:0/14:0/14:0/14:0)	736.7	279.2	-50	100
DHA	Arachidonic acid-d8_MRM	327.2	229.1	-19	0.4
EPA	Arachidonic acid-d8_MRM	301.3	257.1	-19	5
FAHFA(18:1/9-O-18:0)	PG(15:0/18:1(d7))	563.6	281	-50	100
GD1a/b(d36:1)	GM3(d18:1/18:0(d5))	917.5	290.1	-65	1.7
GD3(d34:1)	GM3(d18:1/18:0(d5))	720.9	290.1	-65	3.1
GD3(d36:1)	GM3(d18:1/18:0(d5))	734.9	290.1	-65	100
GM3(d18:1/18:0(d5))	IS	1184.8	290.1	-65	NA
GM3(d34:1)	GM3(d18:1/18:0(d5))	1151.7	290.1	-65	0
GM3(d36:1)	GM3(d18:1/18:0(d5))	1179.8	290.1	-65	0
GQ1b(d36:1)	GM3(d18:1/18:0(d5))	1208.6	290.1	-65	100
Hemi-BMP(14:0/14:0)_14:0	IS	875.5	227.3	-50	NA

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

Hemi-BMP(18:1/18:1)_16:0	Hemi-BMP(14:0/14:0)_14:0	1011.7	281.3	-50	100
Hemi-BMP(18:1/18:1)_18:0	Hemi-BMP(14:0/14:0)_14:0	1039.7	281.3	-50	100
Hemi-BMP(18:1/18:1)_18:1	Hemi-BMP(14:0/14:0)_14:0	1037.7	281.3	-50	100
Hemi-BMP(20:4/20:4)_16:0	Hemi-BMP(14:0/14:0)_14:0	1056.8	303.3	-50	100
Hemi-BMP(20:4/20:4)_18:0	Hemi-BMP(14:0/14:0)_14:0	1185.8	303.3	-50	100
Hemi-BMP(20:4/20:4)_18:1	Hemi-BMP(14:0/14:0)_14:0	1183.8	303.3	-50	100
Hemi-BMP(20:4/20:4)_20:4	Hemi-BMP(14:0/14:0)_14:0	1104.8	303.3	-50	100
Hemi-BMP(22:6/22:6)_16:0	Hemi-BMP(14:0/14:0)_14:0	1103.7	327.3	-50	100
Hemi-BMP(22:6/22:6)_18:0	Hemi-BMP(14:0/14:0)_14:0	1131.7	327.3	-50	100
Hemi-BMP(22:6/22:6)_18:1	Hemi-BMP(14:0/14:0)_14:0	1129.7	327.3	-50	100
Hemi-BMP(22:6/22:6)_22:6	Hemi-BMP(14:0/14:0)_14:0	1175.7	327.3	-50	100
Linoleic acid	Arachidonic acid-d8	279.2	279.2	-38	2.1
Linolenic acid	Arachidonic acid-d8	277.2	277.2	-10	100
LPE(16:0)	LPE(18:1(d7))	452.2	255.3	-50	0
LPE(18:0)	LPE(18:1(d7))	480.31	283.3	-50	0
LPE(18:1(d7))	IS	485.3	288.3	-50	NA
LPG(16:0)	LPE(18:1(d7))	483.3	255.3	-50	18.4
LPG(18:0)	LPE(18:1(d7))	511.3	283.3	-50	12.7
LPG(18:1)	LPE(18:1(d7))	509.3	281.3	-50	0.1
LPG(20:4)	LPE(18:1(d7))	531.3	303.3	-50	100
LPG(22:6)	LPE(18:1(d7))	555.3	327.3	-50	100
LPI(16:0)	LPE(18:1(d7))	571.3	241.1	-50	0.1
LPI(18:0)	LPE(18:1(d7))	599.3	241.1	-50	0
Oleic acid	Arachidonic acid-d8	281.2	281.2	-10	0.1
PA(15:0/18:1(d7))	IS	666.52	241.3	-50	NA
PA(16:0_18:1)	PA(15:0/18:1(d7))	673.5	255.3	-50	0.2
PA(18:0_18:1)	PA(15:0/18:1(d7))	701.5	283.3	-50	4.5
PA(18:0_20:4)	PA(15:0/18:1(d7))	723.5	283.3	-50	100
PA(18:0_22:6)	PA(15:0/18:1(d7))	747.5	283.3	-50	100
PA(18:1/18:1)	PA(15:0/18:1(d7))	699.5	281.3	-50	15.1
Palmitic acid	Arachidonic acid-d8	255.1	255.1	-10	0
Palmitoleic acid	Arachidonic acid-d8	253.1	253.1	-10	0.7

PE(15:0/18:1(d7))	IS	709.6	241.3	-50	NA
PE(O-16:0/20:4)	PE(15:0/18:1(d7))	724.5	303.2	-50	0
PE(O-16:0/22:6)	PE(15:0/18:1(d7))	748.5	327.2	-50	0
PE(O-18:0/20:4)	PE(15:0/18:1(d7))	752.6	303.2	-50	0
PE(O-18:0/22:6)	PE(15:0/18:1(d7))	776.6	327.2	-50	0
PE(P-16:0/20:4)	PE(15:0/18:1(d7))	722.6	303.3	-50	0
PE(P-16:0/20:5)	PE(15:0/18:1(d7))	720.6	301.3	-50	0
PE(P-16:0/22:4)	PE(15:0/18:1(d7))	750.6	331.3	-50	0
PE(P-16:0/22:6)	PE(15:0/18:1(d7))	746.6	327.3	-50	0
PE(P-18:0/18:1)	PE(15:0/18:1(d7))	728.6	281.3	-50	0
PE(P-18:0/18:2)	PE(15:0/18:1(d7))	726.6	279.2	-50	0
PE(P-18:0/20:4)	PE(15:0/18:1(d7))	750.6	303.3	-50	0
PE(P-18:0/20:5)	PE(15:0/18:1(d7))	748.6	301.3	-50	0
PE(P-18:0/22:6)	PE(15:0/18:1(d7))	774.6	327.3	-50	0
PE(P-18:1/20:4)	PE(15:0/18:1(d7))	748.5	303.3	-50	0
PE(P-18:1/22:6)	PE(15:0/18:1(d7))	772.5	327.3	-50	0
PEth(16:0_18:1)	PE(15:0/18:1(d7))	772.5	255.1	-50	0.2
PEth(18:1/18:1)	PE(15:0/18:1(d7))	773.6	281.2	-50	100
PG(15:0/18:1(d7))	IS	740.55	241.3	-50	NA
PG(16:0_18:1)	PG(15:0/18:1(d7))	747.5	255.3	-50	0
PG(16:0_20:4)	PG(15:0/18:1(d7))	769.5	255.3	-50	47.5
PG(16:0_22:6)	PG(15:0/18:1(d7))	795.5	255.3	-50	100
PG(18:0_18:1)	PG(15:0/18:1(d7))	775.5	281.3	-50	0
PG(18:0_20:4)	PG(15:0/18:1(d7))	797.5	283.3	-50	4
PG(18:0_22:6)	PG(15:0/18:1(d7))	823.5	283.3	-50	100
PG(18:1/18:1)	PG(15:0/18:1(d7))	773.4	281.4	-50	0.1
PI(15:0/18:1(d7))	IS	828.6	241.3	-50	NA
PI(16:0_18:1)	PI(15:0/18:1(d7))	835.6	255.3	-50	0
PI(16:0_20:4)	PI(15:0/18:1(d7))	857.6	255.3	-50	0
PI(16:0_22:6)	PI(15:0/18:1(d7))	881.6	255.3	-50	0
PI(18:0_18:1)	PI(15:0/18:1(d7))	863.6	283.3	-50	0
PI(18:0_20:4)	PI(15:0/18:1(d7))	885.6	283.3	-50	0

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

PI(18:0_22:6)	PI(15:0/18:1(d7))	909.6	283.3	-50	0
PI(18:1/18:1)	PI(15:0/18:1(d7))	861.6	281.3	-50	0
PI(20:4/20:4)	PI(15:0/18:1(d7))	905.6	303.3	-50	100
PS(15:0/18:1(d7))	IS	753.55	241.3	-50	NA
PS(16:0_18:1)	PS(15:0/18:1(d7))	760.6	255.3	-50	66.8
PS(16:0_20:4)	PS(15:0/18:1(d7))	782.6	255.3	-50	100
PS(16:0_22:6)	PS(15:0/18:1(d7))	806.6	255.3	-50	100
PS(18:0_18:1)	PS(15:0/18:1(d7))	788.6	283.3	-50	100
PS(18:0_20:4)	PS(15:0/18:1(d7))	810.6	283.3	-50	100
PS(18:0_22:6)	PS(15:0/18:1(d7))	834.6	283.3	-50	100
PS(18:1/18:1)	PS(15:0/18:1(d7))	786.6	281.3	-50	66.6
PS(22:6/22:6)	PS(15:0/18:1(d7))	878.5	327.3	-50	100
Stearic acid	Arachidonic acid-d8	283.2	283.2	-10	0

Note: IS - Internal standard;

The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples.

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

LCMS targeted analysis of metabolites in plasma

Metabolite analyses were performed on liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+ or TQ 6495C).

For positive ionization mode, 5 µL of sample was injected on a BEH amide 1.7 µm, 2.1×150 mm column (Waters Corporation, Milford, Massachusetts, USA) using a flow rate of 0.40 mL/min at 40°C. Mobile phase A consisted of water with 10 mM ammonium formate + 0.1% formic acid. Mobile phase B consisted of acetonitrile with 0.1% formic acid. The gradient was programmed as follows: 0.0–1.0 min at 95% B; 1.0–7.0 min to 50% B; 7.0–7.1 min to 95% B; and 7.1–10.0 min at 95% B. Electrospray ionization was performed in positive ion mode. For the QTRAP 6500+ we applied the following settings: curtain gas at 30 psi; collision gas was set at medium; ion spray voltage at 5500 V; temperature at 600°C; ion source Gas 1 at 50 psi; ion source Gas 2 at 60 psi; entrance potential at 10 V; and collision cell exit potential at 12.5 V. Quantification was performed using MultiQuant 3.02 (Sciex).

For negative ionization mode, 5 µL of sample was injected on an Agilent InfinityLab Poroshell 120 HILIC-Z P 2.7 µm, 2.1×50 mm (Agilent Technologies Inc.); using a flow rate of 0.5 mL/min at 40°C. Mobile phase A consisted of water with 10 mM ammonium acetate, pH 9. Mobile phase B consisted of acetonitrile:water 9:1 with 10 mM ammonium acetate, pH 9. The gradient was programmed as follows: 0.0–1.0 min at 95% B; 1.0–6.0 min to 50% B; 6.0–6.5 min to 95% B; and 6.5–10.0 min at 95% B. Electrospray ionization was performed in negative ion mode. For the Agilent TQ 6495C we applied the following settings: gas temp at 150°C; gas flow 17 l/min; nebulizer 22 psi; sheath gas temp 380°C; sheath gas flow 12 l/min; capillary 2500 V; nozzle voltage 500 V. Quantification was performed using Skyline (v19.1;University of Washington).

Data acquisition was performed in multiple reaction monitoring mode (MRM) with the collision energy (CE) values reported in Tables 3 and 4. Metabolites were quantified using a mixture of non-endogenous internal standards as reported in Table 3 and 4.

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

Table 3: Metabolomics in positive mode parameters

Name	Internal Standard	Q1 m/z	Q3 m/z	CE (V)	% Samples with missing area ratio (Plasma)
1-Methylhistidine	Arginine-d4C13	170	124	20	1.3
1-Methylnicotinamide	15N4-Inosine	137	94	20	0.6
1-Methylxanthine	13C5-Hypoxanthine	167	110	30	5.3
1-Methylxanthosine	13C5-Hypoxanthine	299	167.1	30	100
1,3-15N2-Uric acid	IS	171.1	143	20	NA
13C10,15N5-Adenosine	IS	283	146.1	27	NA
13C3-Thiamine	IS	268	122.07	25	NA
13C5-Hypoxanthine	IS	142.2	124	20	NA
13C5-Xanthosine	IS	290	153	20	NA
15N4-Inosine	IS	273.1	141.01	20	NA
3-Hydroxy-N6,N6,N6-Trimethyllysine	Arginine-d4C13	205.16	128.1	25	17.2
3-Hydroxybutyric acid	Methionine-d3	105	87	20	16.9
3-Hydroxykynurenine	Niacinamide-d4	225.1	179.1	15	0.1
3-Methoxytyrosine	Methionine-d3	212.1	166.1	20	6.8
3-Methylxanthine	13C5-Hypoxanthine	167	69	30	100
3-Methylxanthosine	13C5-Hypoxanthine	299.1	167.1	30	100
4-Hydroxyproline	6C13-Tyrosine	132	68.1	20	0
4-Trimethylammoniobutanal	Arginine-d4C13	130.12	71.05	20	0.1
5-Glutamylalanine	6C13-Phenylalanine	219.1	90.1	12	100
5-Hydroxyindoleacetic acid	Niacinamide-d4	192.07	146.1	20	1
5-Hydroxytryptophan	Methionine-d3	221.11	175	20	2.1
5'-Methylthioadenosine	13C10,15N5-Adenosine	298	136	20	0.1
6-Aminouracil	15N4-Inosine	128	85	19	100
6C13-Phenylalanine	IS	172.1	126.1	18	NA
6C13-Tyrosine	IS	188.1	142.1	24	NA
7-alpha-Hydroxy-3-oxo-4-cholestenoate	Niacinamide-d4	431.32	395.3	15	0.2
7-Methylguanine	15N4-Inosine	166.11	124.1	25	100

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

7-Methylguanosine	13C10,15N5-Adenosine	298	166	20	100
7-Methylxanthine	13C5-Hypoxanthine	167.1	69	30	16.6
7-Methylxanthosine	13C5-Hypoxanthine	299.2	167.1	30	100
8-Hydroxy-deoxyguanosine	15N4-Inosine	284.1	140	20	1.6
8-Hydroxyguanosine	15N4-Inosine	300	168	20	0
8-Oxo-deoxyguanosine-13C,15N2	IS	287.3	171	20	NA
9-Hexadecenoylcarnitine	Palmitoylcarnitine-d3	398.3	85	35	0
Acetylcarnitine	Acetylcarnitine-d3	204.1	85	25	0
Acetylcarnitine-d3	IS	207.14	85	25	NA
Adenine	13C5-Hypoxanthine	136.1	119	30	0
Adenosine	13C10,15N5-Adenosine	268.15	136.1	27	62.7
Adenosine diphosphate ribose	Arginine-d4C13	560.1	136.1	30	100
Adenosylcobalamin	6C13-Phenylalanine	801.5	676.3	31	100
AICA-riboside	15N4-Inosine	259	110	20	100
Alanine	Alanine-d4	90.1	44.2	20	0
Alanine-d4	IS	94.08	48.1	25	NA
alpha-Tocopherol	6C13-Phenylalanine	431.4	165.1	20	0
Arabitol	6C13-Phenylalanine	153.1	153.1	10	0
Arginine	Arginine-d4C13	175.12	70	27	0.1
Arginine-d4C13	IS	180.1	75	27	NA
Ascorbate	13C5-Hypoxanthine	177	95	20	27.1
Asparagine	N15C13-Glycine	133.06	74.02	15	0
Aspartic acid	N15C13-Glycine	134.04	74.02	18	17.1
Aspartic acid-d3	IS	137.04	75.02	18	NA
Asymmetric dimethylarginine	N15C13-Glycine	203.15	70.3	40	0
Betaine	Betaine-d11	118.1	59.1	20	0
Betaine-d11	IS	129.1	68.1	20	NA
Bilirubin	Niacinamide-d4	585.3	299.1	20	2.6
Butyrobetaine	Carnitine-d9	146.1	87	20	0
Butyrylcarnitine	Butyrylcarnitine-d3	232.2	85	30	0
Butyrylcarnitine-d3	IS	235.16	85	30	NA
Cadaverine	6C13-Phenylalanine	102.9	86.1	14	0.6

cADPRibose	Arginine-d4C13	542.1	136.1	30	100
Caffeine	Niacinamide-d4	195.1	138	20	0.1
Capsaicin	Niacinamide-d4	306.2	137.1	20	100
Carnitine	Carnitine-d9	162.1	103.05	20	0
Carnitine-d9	IS	171.16	85	20	NA
Choline	Acetylcarnitine-d3	104	60	21	0
Citrulline	Citrulline-d2	176.1	113.1	12	0
Citrulline-d2	IS	178.11	115.1	12	NA
Cotinine	Niacinamide-d4	177.1	80	32	52.5
Creatine	Alanine-d4	132	90	18	0
Creatinine	15N4-Inosine	114	44.1	22	0
Curcumin	Niacinamide-d4	369	177	30	0
Cyanocobalamin	6C13-Phenylalanine	678.5	997.5	28	100
Cystathionine	6C13-Phenylalanine	223.1	134	15	13.3
Cysteine	Alanine-d4	122.03	76.02	15	100
Cysteinylglycine	6C13-Phenylalanine	179.1	162.2	15	0
Cystine	Arginine-d4C13	241	74	25	0.1
Cytosine	6C13-Phenylalanine	112.1	94.9	25	100
Decanoylcarnitine	Octanoylcarnitine-d3	316.2	85	35	0
Demethoxycurcumin	6C13-Phenylalanine	339.1	255.2	25	100
Deoxyadenosine	13C5-Hypoxanthine	252.1	136	20	100
Deoxyguanosine	15N4-Inosine	268.1	152	20	100
Deoxyinosine	13C5-Hypoxanthine	253	137	20	100
Dimethylethanolamine	15N4-Inosine	90	72	15	0
Dimethylglycine	Alanine-d4	104.02	58	21	0
Dodecanoylcarnitine	Isovalerylcarnitine-d3	343.3	85	30	87.7
Dopa	Alanine-d4	198	152	20	31.2
Dopamine	15N4-Inosine	154	137	13	100
Dopamine 3-o-sulfate	13C5-Hypoxanthine	234	137.1	15	19.9
Dopamine 4-o-sulfate	13C5-Hypoxanthine	234.1	137.2	15	33
Epinephrine	6C13-Phenylalanine	166	107	20	0
Ergothioneine	Methionine-d3	230.1	127	22	10.3

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

Ethanolamine	Methionine-d3	62	44.2	12	0.2
Formylanthranilic acid	Niacinamide-d4	166	120	16	100
gamma-Aminobutyric acid	6C13-Phenylalanine	104.1	87	13	100
Gamma-Glutamylcysteine	6C13-Phenylalanine	251.1	122.02	15	100
Glucosamine	Glutamic acid-d3	180	162	20	100
Glucose	6C13-Phenylalanine	202.8	202.8	10	0
Glutamic acid	Glutamic acid-d3	148.06	84.04	20	0
Glutamic acid-d3	IS	151	133.1	20	NA
Glutamine	Glutamic acid-d3	147.08	84	15	0
Glutathione	Glutamic acid-d3	308.1	179.1	17	23.5
Glutathione disulfide	Arginine-d4C13	613.2	355.1	25	100
Glutathione-(glycine-13C2,15N)	IS	311.3	182.1	16	NA
Glycerophosphocholine	N15C13-Glycine	258.1	104	16	0
Glycine	N15C13-Glycine	76.04	30	25	0
Guanine	15N4-Inosine	152.2	110	20	87.6
Guanosine	15N4-Inosine	284.1	152	20	100
Hexanoylcarnitine	Propionylcartinine-d3	260.2	85	35	0
Histamine	N15C13-Glycine	112.09	95	20	16.3
Histidine	Arginine-d4C13	156.08	110.07	16	0.2
Homocysteine	Valine-d8	136.12	90.1	17	0
Homoserine	N15C13-Glycine	120.15	56.2	24	0
Hydroxybutyrylcarnitine	Carnitine-d9	248.2	85	30	0
Hydroxyhexanoylcarnitine	Carnitine-d9	276.2	85	30	0
Hydroxyisovaleroyl carnitine	Carnitine-d9	262.2	85	30	0
Hydroxypropionylcarnitine	Propionylcartinine-d3	234.1	85	30	100
Hypoxanthine	13C5-Hypoxanthine	137.2	118.8	20	0.2
Imidazoleacetic acid	Alanine-d4	127	81	20	0
Indole	Niacinamide-d4	118	91	20	2.1
Inosine	15N4-Inosine	269.1	137.01	20	8.6
Isovalerylcarnitine	Propionylcartinine-d3	246.2	85	33	0
Isovalerylcarnitine-d3	IS	255.2	85	30	NA
Kynurenic acid	15N4-Inosine	190.05	116	36	29.3

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

Kynurenine	6C13-Phenylalanine	209	146	25	0
Leucine	Leucine-d3	132.1	86.01	15	0
Leucine-d3	IS	135.1	89.1	15	NA
Linoleyl carnitine	Palmitoylcarnitine-d3	424.3	85	45	0
LPC(18:1(d7))	IS	529.3	184.1	40	NA
Lysine	Arginine-d4C13	147.11	84	20	0
Mannitol	6C13-Phenylalanine	183	69	20	100
Mannose	6C13-Phenylalanine	202.81	202.81	15	0
Melatonin	15N4-Inosine	233.1	174.1	20	0.4
Methionine	Methionine-d3	150.06	104.05	10	0
Methionine S-oxide	N15C13-Glycine	166.05	74.02	20	0
Methionine-d3	IS	153.07	107	20	NA
Methylcobalamin	6C13-Phenylalanine	673	971.5	42	100
Myristoylcarnitine	Myristoylcarnitine-d9	372	85	40	2.8
Myristoylcarnitine-d9	IS	381.36	85	35	NA
N-Acetylalanine	Octanoylcarnitine-d3	160.1	72.1	18	100
N-Acetylaspartic acid	13C5-Hypoxanthine	176.06	176.06	10	100
N-Acetylcysteine	6C13-Phenylalanine	163	121.2	15	100
N-Acetylglutamic acid	Glutamic acid-d3	189	130	20	100
N-Acetylneuraminic acid	N15C13-Glycine	310.1	274	15	100
N-Acetylputrescine	Leucine-d3	131.1	114	12	100
N-Acetylserine	6C13-Phenylalanine	148	106	14	0.3
N-Alpha-acetyllysine	Alanine-d4	189	84	18	9.8
N,N'-Bis(gamma-glutamyl)cystine	6C13-Phenylalanine	499.1	453.1	15	100
N1-Acetylspermidine	N15C13-Glycine	188.2	171	19	100
N1-Acetylspermine	Arginine-d4C13	245.2	100.1	22	100
N1-N12-Diacetylspermine	N15C13-Glycine	287.2	100.1	22	100
N1-N8-Diacetylspermidine	N15C13-Glycine	230.5	100.2	19	0
N1-N8-Diacetylspermidine-d6	IS	236.2	103.2	19	NA
N15C13-Glycine	IS	78.04	32	25	NA
N6,N6,N6-Trimethyllysine	Arginine-d4C13	189.16	84.08	20	0.1
NAD	6C13-Phenylalanine	664.1	428.2	24	100

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Niacinamide	Niacinamide-d4	123.06	80	24	0
Niacinamide-d4	IS	127	84	24	NA
Nicotinamide mononucleotide	Valine-d8	335.2	123.1	30	100
Nicotinamide riboside	Alanine-d4	255	123	30	61.2
Nicotinic acid	Niacinamide-d4	124.1	78	30	0
Nitrotyrosine	6C13-Phenylalanine	227.1	210	12	100
Nonanoylcarnitine	Isovalerylcarnitine-d3	302.2	85	30	0.2
Norepinephrine	6C13-Phenylalanine	152	107	15	100
Octanoylcarnitine	Octanoylcarnitine-d3	288.2	85	33	0
Octanoylcarnitine-d3	IS	291.2	85	33	NA
Ornithine	Ornithine-d2	133	70	15	0
Ornithine-d2	IS	135.1	117.2	15	NA
Palmitoylcarnitine	Palmitoylcarnitine-d3	400	85	40	0.1
Palmitoylcarnitine-d3	IS	403.35	85	40	NA
Paraxanthine	Niacinamide-d4	181.1	124	24	0.1
Phenylalanine	6C13-Phenylalanine	166.09	120.08	18	0
Piperanine	Niacinamide-d4	288.2	135.2	33	100
Piperine	Niacinamide-d4	286.1	201.1	20	0.1
Pipernonaline	Niacinamide-d4	342.1	229.2	23	100
Piperyline	Niacinamide-d4	272.2	201.1	28	100
Proline	6C13-Tyrosine	116	70.1	20	0
Propionylcarnitine	Propionylcartinine-d3	218.1	85	20	0
Propionylcartinine-d3	IS	221.15	85	20	NA
Putrescine	N15C13-Glycine	89	72	20	57.7
Pyridoxamine	Alanine-d4	169	152	20	0
Pyroglutamic acid	Octanoylcarnitine-d3	130	84	15	0
S-Adenosylhomocysteine	N15C13-Glycine	385.1	136	21	46.7
S-Adenosylmethionine	Arginine-d4C13	399	250	20	100
Sarcosine	Alanine-d4	90.04	44.1	20	0.1
Serine	N15C13-Glycine	106	60	16	0
Serotonin	15N4-Inosine	177	160	15	0.2
Spermidine	Arginine-d4C13	146.16	72	17	100

Spermine	6C13-Phenylalanine	203.1	129.1	15	100
Stearoylcarnitine	Isovalerylcarnitine-d3	428.4	85	35	0
Sucrose	6C13-Phenylalanine	364.8	202.8	35	0.4
Taurine	6C13-Tyrosine	126.02	108	15	0
Tetrahydrocurcumin	6C13-Phenylalanine	355	137.1	24	100
Theobromine	Niacinamide-d4	181	138.3	25	100
Theophylline	Niacinamide-d4	181.13	124	20	100
Thiamine	13C3-Thiamine	265	122	25	5.3
Threonine	N15C13-Glycine	120	74	13	0
Trigonelline	Methionine-d3	138.1	94.1	20	0
Trimethylamine-d9 N-oxide	IS	85	66	20	NA
Trimethylamine-N-oxide	Trimethylamine-d9 N-oxide	76.08	58.07	20	0
Tryptophan	6C13-Phenylalanine	205.01	146.06	20	0
Tyrosine	6C13-Phenylalanine	182.1	136	15	0
Uracil	Palmitoylcarnitine-d3	113	70	23	100
Ureidopropionic acid	Palmitoylcarnitine-d3	133	115	20	100
Uric acid	1,3-15N2-Uric acid	169.1	141	20	0.2
Uridine	13C5-Hypoxanthine	245	113	18	0
Valerobetaine	Valerobetaine-d9	160	101.1	21	0
Valerobetaine-d9	IS	169.13	101.1	21	NA
Valine	6C13-Phenylalanine	118.09	55	20	0
Valine-d8	IS	126	80.1	20	NA
Xanthine	15N4-Inosine	153	110	20	0.6
Xanthosine	15N4-Inosine	285	153	20	8.9

Note: IS - Internal standard;

The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples.

Table 4: Metabolomics in negative mode parameters

Name	**Internal Standard**	**Q1 m/z**	**Q3 m/z**	**CE (V)**	**% Samples with missing area ratio (Plasma)**
13C3-Citric acid	IS	194.2	113.0	10	NA

13C4-Fumaric acid	IS	119.2	74.3	5	NA
13C4-Malic acid	IS	137.2	119.2	5	NA
13C4-Succinic acid	IS	121.3	76.0	10	NA
13C5-Oxoglutaric acid	IS	150.2	105.3	5	NA
13C6-Benzoic acid	IS	127.0	83.0	10	NA
13C8-Octanic acid	IS	151.1	151.1	5	NA
15N5-ADP	IS	431.1	79	50	NA
2-Hydroxyglutarate	Glutamic acid-d3	145	101	5	100
2-Phosphoglyceric acid	Glutamic acid-d3	185.5	97.0	20	100
3-Hydroxybutyric acid	Glutamic acid-d3	103.2	59.2	5	16.9
3-Phosphoglyceric acid	Glutamic acid-d3	185.5	97.1	15	100
6-Phosphogluconic acid	Glutamic acid-d3	275.0	97.0	20	100
Acetoacetic acid	Glutamic acid-d3	101	101	5	100
Adenosine monophosphate	U-13C10,U-15N5-Adenosine monophosphate	346	79	30	100
Adenosine triphosphate	Adenosine triphosphate-d4	506.1	158.9	40	100
Adenosine triphosphate-d4	IS	510	158.9	40	NA
ADP	15N5-ADP	426.1	79	50	100
Alanine	Alanine-d4	88	88	5	0
Alanine-d4	IS	92	92	5	NA
Arginine	Arginine-d4C13	173	131	15	0.1
Arginine-d4C13	IS	178	136	15	NA
Asparagine	Glutamic acid-d3	131	114	15	0
Aspartic acid	Aspartic acid-d3	132	88	10	17.1

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Aspartic acid-d3	IS	135	91	10	NA
cis-Aconitic acid	Glutamic acid-d3	173.0	129.0	5	0
Dihydroxyacetone phosphate	Glutamic acid-d3	169.1	79.0	50	100
Erythose 4-phosphate	Glutamic acid-d3	199.0	97.0	30	12.7
Fructose 1-phosphate	Glutamic acid-d3	259.1	79.0	30	100
Fructose 1,6-bisphosphate	Glutamic acid-d3	339.0	97.0	50	100
Fructose 6-phosphate	Glutamic acid-d3	259.0	97.1	30	100
Fumaric acid	13C4-Succinic acid	115.2	71.3	5	100
Glucose 1-phosphate	13C5-Oxoglutaric acid	259	241	5	100
Glucose 6-phosphate	13C8-Octanic acid	259.0	97.0	20	1.7
Glutamic acid	Glutamic acid-d3	146	128	10	0
Glutamic acid-d3	Glutamic acid-d3	149	131	10	100
Glutamine	Glutamic acid-d3	145	127	10	0
Glyceraldehyde 3-phosphate	Glutamic acid-d3	169.1	97.0	20	100
Lactic acid	Glutamic acid-d3	89.2	43.2	10	0
Malic acid	13C4-Succinic acid	133.2	115.2	10	100
NAD	Glutamic acid-d3	662.1	540.3	5	100
NADH	Glutamic acid-d3	664.1	408.1	15	100
NADP	Glutamic acid-d3	742.1	620	15	100
NADPH	Glutamic acid-d3	744.1	408	40	100
Oxalacetic acid	13C5-Oxoglutaric acid	131.0	131.0	5	0.3
Oxoglutaric acid	13C5-Oxoglutaric acid	145.2	101.3	5	16.9
Phosphocreatine	Glutamic acid-d3	210	79	40	100

Phosphoenolpyruvic acid	Glutamic acid-d3	167.0	79.0	30	100
Pyruvic acid	13C8-Octanic acid	87.2	43.2	5	100
Ribose 5-phosphate	Glutamic acid-d3	229.0	97.0	30	100
Ribulose 5-phosphate	Glutamic acid-d3	229.0	97.1	30	100
Sedoheptulose 7-phosphate	Glutamic acid-d3	289.0	97.0	20	100
Serine	Glutamic acid-d3	104	74	10	0
Succinic acid	13C4-Succinic acid	117.2	73.2	**10**	0
Sum of Citric acid and Isocitric acid	Glutamic acid-d3	191.2	111.0	**10**	100
U-13C10,U-15N5-Adenosine monophosphate	IS	361	79	**40**	NA
U-13C3-Pyruvic acid	IS	90.2	45.2	5	NA
U-13C6-Glucose 6-phosphate	IS	265.0	97.0	20	NA
Xylulose 5-phosphate	Glutamic acid-d3	229.1	97.0	30	100

Note: IS - Internal standard;

The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples.

LCMS targeted analysis of GlcCer and GalCer in plasma

Glycosphingolipid analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (TQ 6495C). For each analysis, 5µL of sample was injected on a HALO HILIC 2.0 µm, 3.0 × 150 mm column (Advanced Materials Technology, PN 91813-701) using a flow rate of 0.48mL/min at 45°C. Mobile phase A consisted of 92.5/5/2.5 ACN/IPA/H2O with 5 mM ammonium formate and 0.5% formic acid. Mobile phase B consisted of 92.5/5/2.5 H2O/IPA/ACN with 5 mM ammonium formate and 0.5% formic acid. The gradient was programmed as follows: 0.0–2 min at 100% B, 2.1 min at 95% B, 4.5 min at 85% B, hold to 6.0 min at 85% B, drop to 0% B at 6.1 min and hold to 7.0 min, ramp back to 100% at 7.1 min and hold to 8.5 min. Electrospray ionization was performed in positive mode. For the Agilent TQ 6495C we applied the following settings: gas temp at 180°C; gas flow 17 l/min; nebulizer 35 psi; sheath gas temp 350°C; sheath gas flow 10 l/min; capillary 3500 V; nozzle voltage 500 V. Glucosylceramide and galactosylceramide were identified based on their retention times and MRM properties of commercially available reference standards (Avanti Polar Lipids, Birmingham, AL, USA). Quantification was performed using Skyline (v19.1; University of Washington).

Table 5: GalCer/GluCer acquisition parameters

Name	Internal Standard	Q1 m/z	Q3 m/z	CE (V)	% Samples with missing area ratio (Plasma)
alpha-GalCer(d18:1/16:0)	GalCer(d18:1/15:0)	700.6	264.3	32	100
alpha-GalCer(d18:1/18:0)	GalCer(d18:1/15:0)	728.6	264.3	32	100
alpha-GalCer(d18:1/20:0)	GalCer(d18:1/15:0)	756.6	264.3	32	100
alpha-GalCer(d18:1/22:0)	GalCer(d18:1/15:0)	784.7	264.3	32	100
alpha-GalCer(d18:1/22:1)	GalCer(d18:1/15:0)	782.7	264.3	32	100
alpha-GalCer(d18:1/24:0)	GalCer(d18:1/15:0)	812.7	264.3	32	100
alpha-GalCer(d18:1/24:1)	GalCer(d18:1/15:0)	810.7	264.3	32	100
alpha-GalCer(d18:2/18:0)	GalCer(d18:1/15:0)	726.6	262.3	32	100
alpha-GalCer(d18:2/20:0)	GalCer(d18:1/15:0)	754.6	262.3	32	100
alpha-GalCer(d18:2/22:0)	GalCer(d18:1/15:0)	782.7	262.3	32	100
Cholesteryl galactoside	GlcCer(d18:1(d5)/18:0)	566.6	369.3	13	100
Cholesteryl glucoside	GlcCer(d18:1(d5)/18:0)	566.6	369.3	13	100
Galactosylsphingosine	Glucosylsphingosine(d5)	462.2	282.3	18	100
GalCer(d18:1/15:0)	IS	686.3	264.3	32	NA
GalCer(d18:1/16:0)	GalCer(d18:1/15:0)	700.6	264.3	32	1
GalCer(d18:1/18:0)	GalCer(d18:1/15:0)	728.6	264.3	32	100
GalCer(d18:1/20:0)	GalCer(d18:1/15:0)	756.6	264.3	32	100
GalCer(d18:1/22:0)	GalCer(d18:1/15:0)	784.7	264.3	32	100
GalCer(d18:1/22:1)	GalCer(d18:1/15:0)	782.7	264.3	32	100
GalCer(d18:1/24:0)	GalCer(d18:1/15:0)	812.7	264.3	32	100
GalCer(d18:1/24:1)	GalCer(d18:1/15:0)	810.7	264.3	32	100
GalCer(d18:2/18:0)	GalCer(d18:1/15:0)	726.6	262.3	32	2.9

GalCer(d18:2/20:0)	GalCer(d18:1/15:0)	754.6	262.3	32	2.3
GalCer(d18:2/22:0)	GalCer(d18:1/15:0)	782.7	262.3	32	100
GlcCer(d18:1(d5)/18:0)	IS	733.6	269.3	32	NA
GlcCer(d18:1/16:0(d3))	IS	703.6	264.3	32	NA
GlcCer(d18:1/16:0)	GlcCer(d18:1/16:0(d3))	700.6	264.3	32	0.2
GlcCer(d18:1/18:0)	GlcCer(d18:1(d5)/18:0)	728.6	264.3	32	0.4
GlcCer(d18:1/20:0)	GlcCer(d18:1(d5)/18:0)	756.6	264.3	32	0.2
GlcCer(d18:1/22:0)	GlcCer(d18:1(d5)/18:0)	784.7	264.3	32	0.2
GlcCer(d18:1/22:1)	GlcCer(d18:1(d5)/18:0)	782.7	264.3	32	0.5
GlcCer(d18:1/24:0)	GlcCer(d18:1(d5)/18:0)	812.7	264.3	32	0.2
GlcCer(d18:1/24:1)	GlcCer(d18:1(d5)/18:0)	810.7	264.3	32	0.2
GlcCer(d18:2/18:0)	GlcCer(d18:1(d5)/18:0)	726.6	262.3	32	0.3
GlcCer(d18:2/20:0)	GlcCer(d18:1(d5)/18:0)	754.6	262.3	32	0.3
GlcCer(d18:2/22:0)	GlcCer(d18:1(d5)/18:0)	782.7	262.3	32	0.1
Glucosylsphingosine	Glucosylsphingosine(d5)	462.2	282.3	18	0.2
Glucosylsphingosine(d5)	IS	467.2	287.3	18	NA
Sitosteryl galactoside	GlcCer(d18:1(d5)/18:0)	594.6	397.4	13	100
Sitosteryl glucoside	GlcCer(d18:1(d5)/18:0)	594.6	397.4	13	100

Note: IS - Internal standard;

The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples.

LCMS untargeted analysis of metabolites in plasma

Untargeted metabolite analyses were performed on liquid chromatography (UHPLC Ultimate3000) coupled to electrospray mass spectrometry (Q-Exactive) controlled by Xcalibur 2.3 software (Thermo Fisher Scientific, Waltham, MA). For positive ionization mode, 10µL of sample was injected on a BEH amide 1.7 µm, 2.1×150 mm column (Waters Corporation, Milford, Massachusetts, USA); using the chromatography conditions and gradient as described in targeted metabolite analysis (positive mode). HESI source conditions were set as follows: ionization mode, positive; sheath gas flow rate, 30AU; auxiliary gas flow rate, 10AU; sweep gas flow rate, 3AU; spray voltage, 3.6 kV, capillary temperature, 320°C, S-lens RF, level 50; aux gas heater temperature, 120°C. Full scan (m/z 60–900) settings were as follows: resolution, 70,000 fwhm; AGC target, 3x106; maximum injection time, 200 ms. ddMS2 of pooled samples were conducted using the following settings: dd-MS2 resolution, 70,000 fwhm; AGC target, 3x106, maximum injection time,100 ms; loop count (TopN), 5; isolation window, 2.0 m/z; (N)CE/stepped NCE 10, 20, 40.

For negative ionization mode, 10 µL of sample was injected on an Agilent InfinityLab Poroshell 120 HILIC-Z P 2.7 µm, 2.1×50 mm (Agilent Technologies Inc.) using the chromatography conditions and gradient as described in targeted metabolite analysis (negative mode). HESI source conditions were set as follows: ionization mode, negative; sheath gas flow rate, 30AU; auxiliary gas flow rate, 10AU; sweep gas flow rate, 3AU; spray voltage, -2.5 kV, capillary temperature, 320°C, S-lens RF, level 55; aux gas heater temperature, 120°C. Full scan (m/z 60–900) settings were as follows: resolution, 70,000 fwhm; AGC target, 3x106; maximum injection time, 200 ms. ddMS2 of pooled samples were conducted using the following settings: dd-MS2 resolution, 17,500 fwhm; AGC target, 1x106, maximum injection time, 100 ms; loop count (TopN), 5; isolation window, 4.0 m/z; (N)CE/stepped NCE 10, 20, 40.

Data reporting

Targeted LC-MS/MS data analysis in plasma

Chromatographically resolved analytes were detected with tandem mass spectrometry and visualized and integrated using the MultiQuant software (version 3.3 ABSciex) or Skyline (v19.1, University of Washington). Minimum peak quality criteria was SN ratio >5 and points across baseline >8. Missing peak areas results were identified and visualized using the naniar r package (version 0.6.1). Missing values were imputed using k-nearest neighbors imputation method using aggregated information generated from 5 closest donor values. Imputations were performed using the VIM r package (version 6.1.1). Following data imputation, areas for endogenous metabolites and lipids were divided by areas of analyte specific spiked surrogate stable isotope internal standards (See method supplement table for specific pairing of endogenous to spiked-in internal standards) to calculate area ratios. Subsequently, inter-batch variances of area ratios were corrected by using the Combat function provided in the sva r package (version 3.38). Batch adjustment was performed on blinded data and as such, no additional variables (covariates) are included in the model, and only a constant (intercept) is used. Relative log expression plots of area ratios were plotted to visualize variations within and across batches. Following correction for between batch effects, imputed values were removed. For each non-imputed quantified analyte, un-normalized peak area (UNITS = area), ratios of endogenous metabolite area to surrogate internal standards (UNITS = area_ratio) and batch-adjusted area ratios (UNITS = adjusted_area_ratio) were reported. Batch-adjusted area ratios are recommended for downstream analysis. However, unnormalized areas and area ratios provided should allow for different normalization schemes to be adopted by end data consumers. Note that analytes with missing values in >70% of the samples were not reported.
Data analysis and reports were generated using R version 4.0.2.

Untargeted LC-MS data analysis in plasma

Data files were extracted from Thermo 'raw' format to mzML using proteowizard msconvert (version-3.0.21334). A table of files and batch number was produced using the file name nomenclature. This was fed into the apLCMS algorithm using the latest update (version 6.6.8) for multi-batch processing (two step hybrid), as described in Liu et al.(3) Briefly, this method performs peak detection on each file within a batch, while also gain accuracy and power by using knowledge from other samples within the batch. Once peak detection is finished retention time alignment is performed. The algorithm finds a medium retention time deviation sample within the batch and uses this sample to align the other samples to this medium sample. When all the batches have been processed a pairwise retention time alignment (similar to the above method) and weak-signal recovery is performed across the batches. The weak-signal recovery adds accuracy to downstream data analysis while also adding alignment across batches.

3). Liu, Q., Walker, D., Uppal, K.et al. Addressing the batch effect issue for LC/MS metabolomics data in data preprocessing. Sci Rep10,13856 (2020). https://doi.org/10.1038/s41598-020-70850-0

CSF Testing Design

Targeted metabolomic/lipidomic and untargeted lipidomic analyses were conducted across eight different batches. Samples in each batch were selected to ensure that independent factors and covariates of interest were evenly distributed among the batches and randomized. Additionally, PPMI created a single external reference pooled healthy control (HC) and a single external pooled PD reference CSF sample, which were aliquoted evenly distributed within and between batches. Each pooled sample was independently extracted and injected 3-6 times per batch, and values from replicate extractions and injections were used to calculate assay CV and inter-batch deviations.

Each extracted CSF sample was sequentially analyzed using the following methods: metabolomics (positive and negative ionization modes), lipidomics (positive and negative ionization modes), GlcCer/GalCer panel (positive ionization), GM panel (negative ionization), BMP panel (negative ionization), Polyamine panel (positive ionization) and untargeted lipidomics (positive and negative ionization modes). Untargeted metabolomic analysis was not performed for CSF samples. Detailed descriptions of the sample extraction, analysis, and data reporting procedures are provided below.

CSF sample preparation for LC/MS analysis

Thawed CSF samples were spun down at 3.5k x g for 10 min at 4°C. CSF (20 µL) samples were transferred to a Waters 96-well V-bottom half deep-well plate and sample extractions were performed with the Agilent Bravo Prep Platform (Agilent Technologies, Santa Clara, CA).

Commented [MS1]: 80 ul (20 ul aliquots x the 4 processing preps that follow) used to date?

Sample Processing for targeted metabolomics:

One hundred µL of a methanol:water (80:20, v/v) extraction buffer containing stable-isotope and surrogate internal standards was added to each sample. The plates were shaken for 5 minutes at 4°C and then incubated for 1 hour at -20°C. Samples were centrifuged at 4,100 x g for 20 minutes at 4°C. For metabolite analysis in negative ionization mode, a 40 µL aliquot of the supernatant was dried under N2 gas for 1 hour and resuspended in 80 µL of acetonitrile/water/formic acid (10/90/0.1%). For metabolite analysis in positive ionization mode, a 50 µL aliquot of the supernatant was dried under N2 gas for 1 hour and resuspended in 100 µL of methanol:water (80:20, v/v).

Sample Processing for targeted polyamine:

Thawed CSF samples were centrifuged at 3.5k x g for 10 minutes at 4°C. A 20 µL aliquot of each CSF sample was then transferred to a 96-well plate and processed using the Agilent Bravo Platform (Agilent Technologies, Santa Clara, CA). Each sample was diluted with 100 µL of a methanol:water (80:20, v/v) extraction buffer containing stable-isotope internal standards. To these samples, 20 µL of 100 mM sodium carbonate and 75 µL of 5% benzoyl chloride were added, followed by a 15-minute incubation at 55°C. The benzoyl chloride derivatization reaction was quenched with the addition of 200 µL of 0.1% formic acid and 200 µL of ethyl acetate. The samples were then shaken for 1 minute at room temperature. From the resulting mixture, 80 µL of the top layer was aspirated into a 96-well plate, dried under nitrogen for 30 minutes, and reconstituted in 100 µL of a 1:1 mixture of 0.1% formic acid and acetonitrile for LC-MS analysis.

Commented [MS2]: = a 200 ul ethyl acetate layer — if it's not miscible with aqueous sol'ns, and less dense than the 415 ul aqueous phase despite its containing ~19% methanol (which in pure form is less dense than and miscible with ethyl acetate)

Sample Processing for targeted lipidomics:

One hundred µL of a butanol:methanol (3:1, v/v) extraction buffer containing stable isotope and surrogate internal standards was added to each sample. The plates were shaken for 5 minutes at 4°C and then incubated for 1 hour at -20°C. Samples were centrifuged at 4,100 x g for 20 minutes at 4°C. A 100 µL aliquot of the supernatant was dried under N2 gas for 1 hour and reconstituted in 75 µL of methanol:water (80:20, v/v) for lipidomics analysis. For the analysis of the GlcCer/GalCer panel, a 25 µL aliquot of the methanol:water reconstituted sample was further dried under N2 gas for 4 hours and resuspended in 50 µL of acetonitrile/isopropanol/water (92.5/5/2.5, v/v/v) containing 5 mM ammonium formate and 0.5% formic acid.

Sample Processing for untargeted lipidomics:

One hundred µL of a butanol:methanol (3:1, v/v) extraction buffer containing stable isotope and surrogate internal standards was added to each sample. The samples were shaken for 5 minutes at 4°C and then incubated for 1 hour at -20°C. Following incubation, the samples were centrifuged at 4,100 x g for 20 minutes at 4°C. A 100 µL aliquot of the supernatant was dried under N2 gas for 1 hour and reconstituted in 200 µL of isopropanol:methanol (80:20, v/v) for lipidomics analysis.

LCMS targeted analysis of lipids in CSF

Lipid analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+). For each analysis, 5 µL of sample was injected on a BEH C18 1.7 µm, 2.1×100 mm column (Waters) using the chromatography conditions, gradient, and acquisition parameters as described in LCMS targeted analysis of lipids in plasma.

LCMS targeted analysis of Ganglioside in CSF

Ganglioside analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+). For each analysis, 5 µL of sample was injected on a BEH C18 1.7 µm, 2.1×100 mm column (Waters) using the chromatography conditions, gradient, and acquisition parameters as described in LCMS targeted analysis of lipids in negative mode for plasma.

Table 6: Ganglioside in negative mode parameters

Name	Q1 m/z	Q3 m/z	CE (V)
GM3(d18:1/18:0(d5))	1184.8	290.1	40
GM3(d34:1)	1151.7	290.1	40
GM3(d36:1)	1179.8	290.1	40
GM3(d38:1)	1207.8	290.1	40
GM3(d40:1)	1235.8	290.1	40
GM2(d18:1/16:0(d9))	1363.8	290.1	40
GM2(d34:1)	1354.8	290.1	40
GM2(d36:1)	1382.8	290.1	40
GM2(d38:1)	1410.7	290.1	40
GM2(d40:1)	1438.7	290.1	40
GM1(d18:1/18:0(d3))	1519.9	290.1	40
GM1(d34:1)	1516.9	290.1	40

GM1(d36:1)	1544.9	290.1	40
GM1(d38:1)	1572.9	290.1	40
GM1(d40:1)	1600.9	290.1	40

Note: IS - Internal standard;
The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples

LCMS targeted analysis of metabolites in CSF

Metabolite analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+). For each analysis, 5µL of sample was injected on the BEH amide 1.7 µm, 2.1×150 mm column (positive ionization mode) and Agilent InfinityLab Poroshell 120 HILIC-Z P 2.7 µm, 2.1×50 mm (negative ionization mode) using the chromatography conditions, gradient, and acquisition parameters as described in LCMS targeted analysis of metabolites in plasma.

LCMS targeted analysis of GlcCer and GalCer in CSF

Glycosphingolipid analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (QTRAP 6500+). For each analysis, 5µL of sample was injected on a HALO HILIC 2.0 µm, 3.0 × 150 mm column using the chromatography conditions, gradient, and acquisition parameters as described in LCMS targeted analysis of GlcCer and GalCer in plasma.

LCMS targeted analysis of polyamines in CSF

Polyamine analyses were performed by liquid chromatography (Agilent Infinity II1290) coupled to electrospray mass spectrometry (TQ6500+). For each analysis, 3 µL of sample was injected on a BEH C18 1.7 µm, 2.1×100 mm column (Waters) using a flow rate of 0.40 mL/min at 55°C. Mobile phase A consisted of water with 0.1% formic acid; mobile phase B consisted of acetonitrile 100%. The gradient was programmed as follows: 0.00-0.20 at 2% B, 0.2-2.5 min at 2% B, 3.5-7 from 46-95% B, 7-7 4 95% B, 7.4-7.5 from 95-2% B, and from 7.5-8.5 at 2% B. Electrospray ionization was performed in positive ion mode. The following settings were applied: curtain gas at 40 psi; collision gas was set at 8 psi; ion spray voltage at 5500 V; temperature at 500°C; ion source Gas 1 at 50 psi; ion source Gas 2 at 60 psi; entrance potential at 10 V; and collision cell exit potential at 15 V. Quantification was performed using MultiQuant 3.02 (Sciex).

Table 7: Polyamines acquisition parameters *in positive mode parameters*

Name	Q1 m/z	Q3 m/z	CE (V)
13C4-Putrescine	301.1	180.2	23
13C4-Spermidine	462.1	105	63
Cadaverine	311.1	190	22
N1-Acetylcadaverine	249.1	105.1	36
N1-Acetylputrescine	235.1	105.1	34
N1-Acetylspermidine	396	274	23
N1-Acetylspermine-d3	560	438	33
N1-N8-Diacetylspermidine	334	171.2	23
N8-Acetylspermidine-d3	399	162	37
Ornithine	341.3	105	38
Putrescine	298	105	35
Spermidine	458.2	162.1	43
Spermine	619.2	105.1	87
Spermine-d8	627.4	162.2	51

Note: IS - Internal standard;
The percent of samples with missing area ratio is calculated across the entire set of samples, including the pooled control samples

Untargeted lipidomic analysis in CSF

Untargeted lipidomic analysis was performed on liquid chromatography (1290 Infinity II HPLC) coupled to electrospray mass spectrometry (Agilent 6546 LC/Q-TOF). For positive ionization mode, 3 μL of sample was injected on an AQUITY UPLC® BEH C18 column (1.7 μm × 2.1 mm × 50 mm) with corresponding VanGuard pre-column (Waters Corporation, Milford, MA, USA) at 50 °C with a flow rate of 0.5 mL/min. Mobile phase A consisted of 60:40 acetonitrile/water (v/v) with 10 mM ammonium formate and 0.1% formic acid; Mobile phase B consisted of 90:8:2 IPA:ACN with 0.1% formic acid and 10 mM ammonium formate. The gradient was programmed as follows: 0.0–1.0 min, 20% B; 1.0–2.5 min, 20–30% B; 2.5–3.5 min, 30–45% B; 3.5–5.0 min, 45–60% B; 5.0–7.0 min, 60–65% B; 7.0–9.0 min, 65–65% B; 9.0–14.0 min, 65–90% B; 14.0–15.0 min, 90–98% B; 15.0–15.25 min, 98–98% B, followed by a 2.25 min column flush and re-equilibration. Source conditions were set as follows: Gas Temp 275 °C, Drying Gas 10 L/min, Nebulizer 50 psi, Sheath Gas Temp 350 °C, Sheath Gas flow 12 L/min, VCap 4250 V, Nozzle Voltage (expt) 0 V, Fragmentor 150 V, Skimmer 65 V, Oct 1 RF Vpp 750 V. Full scan (m/z 140 – 1700) settings were as follows: acquisition rate, 2 spectra/s, collision energy, 0 V. Auto MS/MS (data dependent acquisition, DDA) of pooled samples were conducted using the following settings: acquisition rate, 5 spectra/s, collision energy, 20 and 35 V.

For negative ionization mode, 5 µL of sample was injected on an AQUITY UPLC® BEH C18 column (1.7 µm × 2.1 mm × 50 mm) with corresponding VanGuard pre-column (Waters Corporation, Milford, MA, USA) at 50 °C with a flow rate of 0.5 mL/min. Mobile phase A consisted of 60:40 acetonitrile/water (v/v) with 10 mM ammonium formate; mobile phase B consisted of 90:8:2 IPA:ACN:water with and 10 mM ammonium formate. The gradient was programmed as follows: 0.0–1.0 min, 20% B; 1.0–2.5 min, 20–30% B; 2.5–3.5 min, 30–45% B; 3.5–5.0 min, 45–60% B; 5.0–7.0 min, 60–65% B; 7.0–9.0 min, 65–65% B; 9.0–14.0 min, 65–90% B; 14.0–15.0 min, 90–98% B; 15.0–15.25 min, 98–98% B, followed by a 2.25 min column flush and re-equilibration. Source conditions were set as follows: Gas Temp 275 °C, Drying Gas 10 L/min, Nebulizer 50 psi, Sheath Gas Temp 350 °C, Sheath Gas flow 12 L/min, VCap 4000 V, Nozzle Voltage (expt) 0 V, Fragmentor 150 V, Skimmer 65 V, Oct 1 RF Vpp 750 V. Full scan (m/z 140 – 1700) settings were as follows: acquisition rate, 2 spectra/s, collision energy, 0 V. Auto MS/MS of pooled samples were conducted using the following settings: acquisition rate, 5 spectra/s, collision energy, 20 and 35 V.

Data Reporting

Targeted LC-MS/MS data analysis in CSF

Chromatographically resolved analytes were detected with tandem mass spectrometry and visualized and integrated using the MultiQuant software (version 3.3 ABSciex). Minimum peak quality criteria was SN ratio >5 and points across baseline >8. Missing peak areas results were identified and visualized using the naniar r package (version 1.1.0). Integrated peak areas of endogenous compounds were normalized to the peak areas of specific internal standards to obtain area ratio data. The specific pairings between endogenous analytes and spiked-in internal standards are provided in Table 8. Values that are missing in 100% of the samples were not reported. For endogenous analytes detected in at least 70% of the samples, additional batch correction was performed. To this end, area ratio values were first log2 transformed and then imputed using the kNN function in the VIM R package (version 6.2.2). Batch correction was performed using the ComBat function from the sva R package (version 3.52.0). Batch adjustment was performed on blinded data and as such, no additional variables (covariates) are included in the model, and only a constant (intercept) is used. For each quantified analyte, unadjusted peak areas (UNITS = area), unadjusted ratios of endogenous metabolite areas to surrogate internal standards (UNITS = area_ratio), and (when determined, as above) batch-adjusted area ratios (UNITS = adjusted_area_ratio) were reported. For each analyte, the primary results are considered to be those reported in units of adjusted_area_ratio when reported, and in units of area_ratio otherwise. Data analysis and reports were generated using R version 4.4.

Commented [MS3]: Would have thought performing/providing batch correction is helpful even for those <70% detected (e.g., for guanosine @58% detected) but there are only a handful <70% but >0% detected, so should be fine to leave as is.

Table 8: Paired surrogate internal standard and corresponding analyte *CSF*

1. ASSAY: METABOLOMICS_POSITIVE

Row	TESTNAME	IS_NAME	% missing
1	1-Methylhistidine	Arginine-d4C13	100.00
2	1-Methylnicotinamide	15N4-Inosine	0.00
3	1-Methylxanthine	13C5-Hypoxanthine	7.50
4	1-Methylxanthosine	13C5-Hypoxanthine	100.00
5	2-Aminobenzoic acid	Niacinamide-d4	100.00
6	3-Hydroxy-N6,N6,N6-Trimethyllysine	Arginine-d4C13	0.00
7	3-Hydroxybutyric acid	Methionine-d3	100.00
8	3-Hydroxykynurenine	Niacinamide-d4	0.16
9	3-Hydroxytyrosol	13C5-Hypoxanthine	100.00
10	3-Methoxytyrosine	Methionine-d3	63.91
11	3-Methylxanthine	13C5-Hypoxanthine	10.00
12	3-Methylxanthosine	13C5-Hypoxanthine	100.00
13	4-Hydroxyproline	13C3-Thiamine	0.00
14	4-Trimethylammoniobutanal	Arginine-d4C13	0.00

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15	5-Glutamylalanine	6C13-Phenylalanine	100.00
16	5-Hydroxyindoleacetic acid	Niacinamide-d4	0.00
17	5-Hydroxytryptophan	Methionine-d3	0.00
18	5'-Methylthioadenosine	13C10,15N5-Adenosine	0.00
19	6-Aminouracil	15N4-Inosine	100.00
20	7-alpha-Hydroxy-3-oxo-4-cholestenoate	Niacinamide-d4	0.00
21	7-Methylguanine	15N4-Inosine	0.00
22	7-Methylguanosine	13C10,15N5-Adenosine	100.00
23	7-Methylxanthine	13C5-Hypoxanthine	7.03
24	7-Methylxanthosine	13C5-Hypoxanthine	100.00
25	8-Hydroxy-deoxyguanosine	15N4-Inosine	14.53
26	8-Hydroxyguanosine	15N4-Inosine	0.00
27	9-Hexadecenoylcarnitine	Palmitoylcarnitine-d3	100.00
28	Acetylcarnitine	Acetylcarnitine-d3	0.00
29	Adenine	13C5-Hypoxanthine	0.00
30	Adenosine	13C10,15N5-Adenosine	0.00
31	Adenosine diphosphate ribose	Arginine-d4C13	100.00
32	Adenosylcobalamin	6C13-Phenylalanine	100.00
33	AICA-riboside	15N4-Inosine	100.00
34	Alanine	Alanine-d4	0.00
35	alpha-Tocopherol	6C13-Phenylalanine	0.00
36	Arabitol	6C13-Phenylalanine	8.28
37	Arginine	Arginine-d4C13	0.00
38	Ascorbate	13C5-Hypoxanthine	100.00
39	Asparagine	N15C13-Glycine	0.00
40	Aspartic acid	Aspartic acid-d3	0.78
41	Asymmetric dimethylarginine	N15C13-Glycine	0.00
42	Betaine	Betaine-d11	0.00
43	Bilirubin	Niacinamide-d4	3.28
44	Butyrobetaine	Carnitine-d9	0.00
45	Butyrylcarnitine	Butyrylcarnitine-d3	0.00

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46	cADPRibose	Arginine-d4C13	100.00
47	Caffeine	Niacinamide-d4	0.00
48	Capsaicin	Niacinamide-d4	100.00
49	Carnitine	Carnitine-d9	0.00
50	Choline	Acetylcarnitine-d3	0.00
51	Citrulline	Citrulline-d2	0.00
52	Cotinine	Niacinamide-d4	100.00
53	Creatine	Alanine-d4	0.00
54	Creatinine	15N4-Inosine	0.00
55	Curcumin	Niacinamide-d4	0.00
56	Cyanocobalamin	6C13-Phenylalanine	100.00
57	Cystathionine	6C13-Phenylalanine	100.00
58	Cysteine	Alanine-d4	100.00
59	Cysteinylglycine	6C13-Phenylalanine	12.97
60	Cystine	Arginine-d4C13	0.16
61	Cytosine	6C13-Phenylalanine	9.53
62	Decanoylcarnitine	Octanoylcarnitine-d3	0.00
63	Demethoxycurcumin	6C13-Phenylalanine	100.00
64	Deoxyadenosine	13C5-Hypoxanthine	87.50
65	Deoxyguanosine	15N4-Inosine	10.47
66	Deoxyinosine	13C5-Hypoxanthine	18.28
67	Dimethylethanolamine	15N4-Inosine	0.00
68	Dimethylglycine	Alanine-d4	0.00
69	Dodecanoylcarnitine	Isovalerylcarnitine-d9	100.00
70	Dopa	Alanine-d4	0.00
71	Dopamine	15N4-Inosine	15.78
72	Dopamine 3-o-sulfate	13C5-Hypoxanthine	4.69
73	Dopamine 4-o-sulfate	13C5-Hypoxanthine	100.00
74	Epinephrine	6C13-Phenylalanine	0.00
75	Ergothioneine	Methionine-d3	0.00
76	Ethanolamine	Methionine-d3	0.00

77	Formylanthranilic acid	Niacinamide-d4	100.00
78	gamma-Aminobutyric acid	6C13-Phenylalanine	0.00
79	Gamma-Glutamylcysteine	6C13-Phenylalanine	75.78
80	Glucosamine	Glutamic acid-d3	100.00
81	Glucose	6C13-Phenylalanine	0.00
82	Glutamic acid	Glutamic acid-d3	0.00
83	Glutamine	Glutamic acid-d3	0.00
84	Glutathione	Glutamic acid-d3	100.00
85	Glutathione disulfide	Arginine-d4C13	100.00
86	Glycerophosphocholine	N15C13-Glycine	0.00
87	Glycine	N15C13-Glycine	0.16
88	Guanine	15N4-Inosine	100.00
89	Guanosine	15N4-Inosine	41.56
90	Hexanoylcarnitine	Propionylcartinine-d3	0.00
91	Histamine	N15C13-Glycine	100.00
92	Histidine	Arginine-d4C13	0.00
93	Homocysteine	Valine-d8	0.00
94	Homoserine	N15C13-Glycine	0.00
95	Hydroxybutyrylcarnitine	Carnitine-d9	100.00
96	Hydroxyhexanoylcarnitine	Carnitine-d9	0.78
97	Hydroxyisovaleroyl carnitine	Carnitine-d9	0.00
98	Hydroxypropionylcarnitine	Propionylcartinine-d3	76.25
99	Hypoxanthine	13C5-Hypoxanthine	0.00
100	Imidazoleacetic acid	Alanine-d4	0.00
101	Indole	Niacinamide-d4	100.00
102	Inosine	15N4-Inosine	0.00
103	Isovalerylcarnitine	Propionylcartinine-d3	0.00
104	Kynurenic acid	15N4-Inosine	1.56
105	Kynurenine	6C13-Phenylalanine	0.00
106	Leucine	Leucine-d3	0.00
107	Linoleyl carnitine	Palmitoylcarnitine-d3	100.00

108	Lysine	Arginine-d4C13	0.00
109	Mannitol	6C13-Phenylalanine	0.00
110	Mannose	6C13-Phenylalanine	0.00
111	Melatonin	15N4-Inosine	0.00
112	Methionine	Methionine-d3	0.00
113	Methionine S-oxide	N15C13-Glycine	0.00
114	Methylcobalamin	6C13-Phenylalanine	100.00
115	Myristoylcarnitine	Myristoylcarnitine-d9	100.00
116	N-Acetylalanine	Octanoylcarnitine-d3	100.00
117	N-Acetylaspartic acid	13C5-Hypoxanthine	100.00
118	N-Acetylcysteine	6C13-Phenylalanine	100.00
119	N-Acetylglutamic acid	Glutamic acid-d3	100.00
120	N-Acetylneuraminic acid	N15C13-Glycine	75.00
121	N-Acetylserine	6C13-Phenylalanine	0.00
122	N-Alpha-acetyllysine	Alanine-d4	0.16
123	N,N'-Bis(gamma-glutamyl)cystine	6C13-Phenylalanine	100.00
124	N1-N12-Diacetylspermine	N15C13-Glycine	12.50
125	N6,N6,N6-Trimethyllysine	Arginine-d4C13	0.00
126	NAD	6C13-Phenylalanine	100.00
127	Niacinamide	Niacinamide-d4	0.00
128	Nicotinamide mononucleotide	Valine-d8	100.00
129	Nicotinamide riboside	Alanine-d4	0.00
130	Nicotinic acid	Niacinamide-d4	100.00
131	Nitrotyrosine	6C13-Phenylalanine	100.00
132	Nonanoylcarnitine	Isovalerylcarnitine-d9	0.00
133	Norepinephrine	6C13-Phenylalanine	100.00
134	Octanoylcarnitine	Octanoylcarnitine-d3	0.00
135	Ornithine	Ornithine-d2	0.00
136	Palmitoylcarnitine	Palmitoylcarnitine-d3	100.00
137	Paraxanthine	Niacinamide-d4	0.00
138	Phenylalanine	6C13-Phenylalanine	0.00

139	Piperanine	Niacinamide-d4	100.00
140	Piperine	Niacinamide-d4	0.78
141	Pipernonaline	Niacinamide-d4	100.00
142	Piperyline	Niacinamide-d4	100.00
143	Proline	13C3-Thiamine	100.00
144	Propionylcarnitine	Propionylcartinine-d3	0.00
145	Pyridoxamine	Alanine-d4	0.00
146	Pyroglutamic acid	Octanoylcarnitine-d3	0.00
147	S-Adenosylhomocysteine	N15C13-Glycine	2.19
148	S-Adenosylmethionine	Arginine-d4C13	12.50
149	Sarcosine	Alanine-d4	0.00
150	Serine	N15C13-Glycine	0.00
151	Serotonin	15N4-Inosine	100.00
152	Stearoylcarnitine	Isovalerylcarnitine-d9	100.00
153	Sucrose	6C13-Phenylalanine	0.00
154	Taurine	13C3-Thiamine	0.00
155	Tetrahydrocurcumin	6C13-Phenylalanine	100.00
156	Theobromine	Niacinamide-d4	0.16
157	Theophylline	Niacinamide-d4	100.00
158	Thiamine	13C3-Thiamine	0.00
159	Threonine	N15C13-Glycine	0.00
160	Trigonelline	Methionine-d3	0.31
161	Trimethylamine-N-oxide	Trimethylamine-d9 N-oxide	0.00
162	Tryptophan	6C13-Phenylalanine	0.00
163	Tyrosine	6C13-Phenylalanine	100.00
164	Uracil	Palmitoylcarnitine-d3	75.00
165	Ureidopropionic acid	Palmitoylcarnitine-d3	2.50
166	Uric acid	1,3-15N2-Uric acid	0.31
167	Uridine	13C5-Hypoxanthine	0.00
168	Valerobetaine	Valerobetaine-d9	0.00
169	Valine	6C13-Phenylalanine	0.00

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170	Xanthine	15N4-Inosine	0.00
171	Xanthosine	15N4-Inosine	4.53

2. ASSAY: METABOLOMICS_NEGATIVE

Row	TESTNAME	IS_NAME	% missing
172	2-Hydroxyglutarate	Glutamic acid-d3	100.00
173	2-Phosphoglyceric acid	Glutamic acid-d3	100.00
174	3-Hydroxybutyric acid	Glutamic acid-d3	0.16
175	3-Phosphoglyceric acid	Glutamic acid-d3	100.00
176	6-Phosphogluconic acid	Glutamic acid-d3	12.66
177	Acetoacetic acid	Glutamic acid-d3	100.00
178	Adenosine monophosphate	U-13C10,U-15N5-Adenosine monophosphate	96.72
179	Adenosine triphosphate	15N5-ADP	100.00
180	ADP	15N5-ADP	96.72
181	Alanine	Arginine-d4C13	100.00
182	Arginine	Arginine-d4C13	100.00
183	Asparagine	Glutamic acid-d3	100.00
184	Aspartic acid	13C4-Succinic acid	100.00
185	cis-Aconitic acid	Glutamic acid-d3	100.00
186	Sum of Citric acid and Isocitric acid	Glutamic acid-d3	0.31
187	Dihydroxyacetone phosphate	Glutamic acid-d3	87.50
188	Erythose 4-phosphate	Glutamic acid-d3	100.00
189	Fructose 1,6-bisphosphate	Glutamic acid-d3	100.00
190	Fructose 6-phosphate	Glutamic acid-d3	100.00
191	Fumaric acid	13C4-Succinic acid	100.00
192	Glucose 1-phosphate	Glutamic acid-d3	100.00
193	Glucose 6-phosphate	Glutamic acid-d3	0.16
194	Glutamic acid	Glutamic acid-d3	100.00
195	Glutamine	Glutamic acid-d3	100.00
196	Glyceraldehyde 3-phosphate	Glutamic acid-d3	100.00
197	Itaconic acid	13C5-Itaconic acid	100.00

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198	Lactic acid	Glutamic acid-d3	0.16
199	Malic acid	13C4-Succinic acid	100.00
200	NAD	Glutamic acid-d3	100.00
201	NADH	Glutamic acid-d3	100.00
202	NADP	Glutamic acid-d3	100.00
203	NADPH	Glutamic acid-d3	100.00
204	Oxalacetic acid	Glutamic acid-d3	100.00
205	Oxoglutaric acid	13C4-Succinic acid	100.00
206	Phosphocreatine	Glutamic acid-d3	100.00
207	Phosphoenolpyruvic acid	Glutamic acid-d3	100.00
208	Pyruvic acid	13C4-Succinic acid	100.00
209	Ribose 5-phosphate	Glutamic acid-d3	0.16
210	Ribulose 5-phosphate	Glutamic acid-d3	100.00
211	Sedoheptulose 7-phosphate	Glutamic acid-d3	100.00
212	Serine	Glutamic acid-d3	100.00
213	Succinic acid	13C4-Succinic acid	0.16
214	Xylulose 5-phosphate	Glutamic acid-d3	100.00
215	Fructose 1-phosphate	Glutamic acid-d3	100.00
216	Glucose 1-phosphate	U-13C6-Glucose 6-phosphate	100.00
217	Glucose 6-phosphate	U-13C6-Glucose 6-phosphate	0.16
218	Oxalacetic acid	U-13C6-Glucose 6-phosphate	100.00

3. ASSAY: POLYAMINE_POSITIVE

Row	TESTNAME	IS_NAME	% missing
219	Cadaverine	13C4-Putrescine	0.00
220	N1-Acetylcadaverine	N1-Acetylspermine-d3	100.00
221	N-Acetylputrescine	N1-Acetylspermine-d3	4.53
222	N1-Acetylspermidine	N8-Acetylspermidine-d3	0.00
223	N1-Acetylspermine	N1-Acetylspermine-d3	100.00
224	N1-N8-Diacetylspermidine	N8-Acetylspermidine-d3	0.00
225	Putrescine	13C4-Putrescine	0.00

226	Spermidine	13C4-Spermidine	0.00
227	Spermine	Spermine-d8	0.00

4. ASSAY: LIPIDOMICS_POSITIVE

Row	TESTNAME	IS_NAME	% missing
228	1-O-Palmitoyl-Cer(d18:1/18:0)	Cer(d18:1/16:0(d7))	100.00
229	24-Hydroxycholesterol	24-Hydroxycholesterol(d7)	100.00
230	3-O-SulfoLacCer(d18:1/18:0)	LacCer(d18:1/17:0)	100.00
231	4-beta-Hydroxycholesterol	Cholesterol(d7)	100.00
232	7-keto-Cholesterol	Cholesterol(d7)	100.00
233	CE HETE	CE(18:1(d7))	100.00
234	CE HODE	CE(18:1(d7))	100.00
235	CE HpODE	CE(18:1(d7))	100.00
236	CE oxoHETE	CE(18:1(d7))	100.00
237	CE oxoODE	CE(18:1(d7))	100.00
238	CE(16:1)	CE(18:1(d7))	0.00
239	CE(18:1)	CE(18:1(d7))	0.00
240	CE(18:2)	CE(18:1(d7))	0.00
241	CE(20:4)	CE(18:1(d7))	0.00
242	CE(20:5)	CE(18:1(d7))	0.00
243	CE(22:6)	CE(18:1(d7))	0.00
244	Cer(d18:0/16:0)	Cer(d18:1/16:0(d7))	100.00
245	Cer(d18:0/18:0)	Cer(d18:1/16:0(d7))	100.00
246	Cer(d18:0/24:0)	Cer(d18:1/16:0(d7))	100.00
247	Cer(d18:0/24:1)	Cer(d18:1/16:0(d7))	100.00
248	Cer(d18:1/16:0)	Cer(d18:1/16:0(d7))	0.16
249	Cer(d18:1/18:0)	Cer(d18:1/16:0(d7))	0.00
250	Cer(d18:1/24:0)	Cer(d18:1/16:0(d7))	0.00
251	Cer(d18:1/24:1)	Cer(d18:1/16:0(d7))	0.16
252	Cholesterol	Cholesterol(d7)	0.16
253	Cholesteryl hexoside	CE(18:1(d7))	100.00

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

254	Coenzyme Q10	TG(15:0/18:1(d7)/15:0)	100.00
255	DG(16:0_18:1)	DG(15:0/18:1(d7))	0.00
256	DG(16:0_20:4)	DG(15:0/18:1(d7))	100.00
257	DG(18:0_18:1)	DG(15:0/18:1(d7))	100.00
258	DG(18:0_20:4)	DG(15:0/18:1(d7))	100.00
259	DG(18:0_22:6)	DG(15:0/18:1(d7))	0.00
260	DG(18:1/18:1)	DG(15:0/18:1(d7))	0.00
261	DG(18:1_20:4)	DG(15:0/18:1(d7))	100.00
262	GB3(d18:1/16:0)	GB3(d18:1/18:0(d3))	100.00
263	GB3(d18:1/18:0)	GB3(d18:1/18:0(d3))	100.00
264	GB3(d18:1/24:0)	GB3(d18:1/18:0(d3))	100.00
265	GB3(d18:1/24:1)	GB3(d18:1/18:0(d3))	100.00
266	HexCer(d18:1/12:0)	GlcCer(d18:1(d5)/18:0)	100.00
267	HexCer(d18:1/16:0)	GlcCer(d18:1(d5)/18:0)	0.00
268	HexCer(d18:1/18:0)	GlcCer(d18:1(d5)/18:0)	0.00
269	HexCer(d18:1/22:0)	GlcCer(d18:1(d5)/18:0)	0.16
270	HexCer(d18:1/24:0)	GlcCer(d18:1(d5)/18:0)	0.16
271	HexCer(d18:1/24:1)	GlcCer(d18:1(d5)/18:0)	0.00
272	Hexosylsphingosine	Glucosylsphingosine(d5)	100.00
273	LacCer(d18:1/16:0)	GlcCer(d18:1(d5)/18:0)	4.69
274	LacCer(d18:1/18:0)	GlcCer(d18:1(d5)/18:0)	0.00
275	LacCer(d18:1/24:0)	GlcCer(d18:1(d5)/18:0)	100.00
276	LacCer(d18:1/24:1)	GlcCer(d18:1(d5)/18:0)	100.00
277	Lactosylsphingosine	Glucosylsphingosine(d5)	100.00
278	LPC(16:0)	LPC(18:1(d7))	0.00
279	LPC(16:1)	LPC(18:1(d7))	0.16
280	LPC(18:0)	LPC(18:1(d7))	0.00
281	LPC(18:1)	LPC(18:1(d7))	0.00
282	LPC(20:4)	LPC(18:1(d7))	0.16
283	LPC(22:6)	LPC(18:1(d7))	0.16
284	LPC(24:0)	LPC(18:1(d7))	100.00

285	LPC(24:1)	LPC(18:1(d7))	100.00
286	LPC(26:0)	LPC(18:1(d7))	100.00
287	LPC(26:1)	LPC(18:1(d7))	100.00
288	lyso-GB3	lyso-GB3-d7	100.00
289	lyso-GB4	lyso-GB3-d7	100.00
290	MG(16:0)	MG(18:1(d7))	87.66
291	MG(16:1)	MG(18:1(d7))	100.00
292	MG(18:0)	MG(18:1(d7))	100.00
293	MG(18:1)	MG(18:1(d7))	100.00
294	MG(20:4)	MG(18:1(d7))	100.00
295	N-Oleoylethanolamine	LPC(18:1(d7))	100.00
296	N-Palmitoyl-O-phosphocholineserine	LPC(18:1(d7))	100.00
297	Palmitoylethanolamine	LPC(18:1(d7))	100.00
298	PC(16:0/5:0(CHO))	PC(15:0/18:1(d7))	87.50
299	PC(16:0/9:0(CHO))	PC(15:0/18:1(d7))	0.00
300	PC(16:0/9:0(COOH))	PC(15:0/18:1(d7))	0.00
301	PC(18:0/20:4(OH[S]))	PC(15:0/18:1(d7))	100.00
302	PC(18:0/20:4(OOH[S]))	PC(15:0/18:1(d7))	100.00
303	PC(34:1)	PC(15:0/18:1(d7))	0.00
304	PC(36:1)	PC(15:0/18:1(d7))	0.00
305	PC(36:2)	PC(15:0/18:1(d7))	0.00
306	PC(36:4)	PC(15:0/18:1(d7))	0.16
307	PC(38:4)	PC(15:0/18:1(d7))	0.00
308	PC(38:6)	PC(15:0/18:1(d7))	0.16
309	PC(40:5)	PC(15:0/18:1(d7))	0.00
310	PC(40:6)	PC(15:0/18:1(d7))	0.16
311	PC(O-16:0/0:0)	LPC(18:1(d7))	0.00
312	PC(O-16:0/2:0)	LPC(18:1(d7))	0.16
313	PC(O-18:0/2:0)	LPC(18:1(d7))	0.16
314	PE(18:0/20:4(OH[S]))	PE(15:0/18:1(d7))	100.00
315	PE(18:0/20:4(OOH[S]))	PE(15:0/18:1(d7))	100.00

316	PE(34:1)	PE(15:0/18:1(d7))	0.00
317	PE(36:1)	PE(15:0/18:1(d7))	0.00
318	PE(36:2)	PE(15:0/18:1(d7))	0.00
319	PE(36:4)	PE(15:0/18:1(d7))	0.16
320	PE(38:4)	PE(15:0/18:1(d7))	0.16
321	PE(38:5)	PE(15:0/18:1(d7))	0.16
322	PE(38:6)	PE(15:0/18:1(d7))	0.16
323	PE(40:4)	PE(15:0/18:1(d7))	0.16
324	PE(40:5)	PE(15:0/18:1(d7))	100.00
325	PE(40:6)	PE(15:0/18:1(d7))	0.00
326	PE(40:7)	PE(15:0/18:1(d7))	0.16
327	Sitosteryl hexoside	CE(18:1(d7))	100.00
328	SM(d18:1/16:0)	SM(d18:1(d9)/18:1)	0.00
329	SM(d18:1/18:0)	SM(d18:1(d9)/18:1)	0.00
330	SM(d18:1/24:0)	SM(d18:1(d9)/18:1)	0.00
331	SM(d18:1/24:1)	SM(d18:1(d9)/18:1)	0.00
332	Sphinganine	Sphingosine(d17:1)	100.00
333	Sphinganine 1-phosphate	Sphingosine 1-phosphate-d7	100.00
334	Sphingosine	Sphingosine(d17:1)	100.00
335	Sphingosine 1-phosphate	Sphingosine 1-phosphate-d7	100.00
336	Sphingosine 1-phosphocholine	LPC(18:1(d7))	100.00
337	TG(18:0_36:2)	TG(15:0/18:1(d7)/15:0)	0.00
338	TG(18:1_34:2)	TG(15:0/18:1(d7)/15:0)	0.00
339	TG(18:1_34:3)	TG(15:0/18:1(d7)/15:0)	50.00
340	TG(20:4_32:1)	TG(15:0/18:1(d7)/15:0)	100.00
341	TG(20:4_34:2)	TG(15:0/18:1(d7)/15:0)	50.00
342	TG(20:4_34:3)	TG(15:0/18:1(d7)/15:0)	100.00
343	TG(20:4_36:0)	TG(15:0/18:1(d7)/15:0)	100.00
344	TG(20:4_36:2)	TG(15:0/18:1(d7)/15:0)	100.00
345	TG(20:4_36:3)	TG(15:0/18:1(d7)/15:0)	100.00
346	TG(22:6_36:2)	TG(15:0/18:1(d7)/15:0)	100.00

347	TG(22:6_38:1)	TG(15:0/18:1(d7)/15:0)	100.00
348	TG(22:6_38:2)	TG(15:0/18:1(d7)/15:0)	100.00

5. ASSAY: LIPIDOMICS_NEGATIVE

Row	TESTNAME	IS_NAME	% missing
349	(3-O-sulfo)GalCer(d18:1/16:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.16
350	(3-O-sulfo)GalCer(d18:1/18:0(2OH)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.00
351	(3-O-sulfo)GalCer(d18:1/18:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.00
352	(3-O-sulfo)GalCer(d18:1/24:0(2OH))	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.16
353	(3-O-sulfo)GalCer(d18:1/24:0)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.00
354	(3-O-sulfo)GalCer(d18:1/24:1(2OH))	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.00
355	(3-O-sulfo)GalCer(d18:1/24:1)	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.00
357	Arachidonic acid_MRM	Arachidonic acid-d8_MRM	0.31
358	BMP(16:0_18:1)	BMP(14:0/14:0)	100.00
359	BMP(16:0_20:4)	BMP(14:0/14:0)	100.00
360	BMP(16:0_22:6)	BMP(14:0/14:0)	100.00
361	BMP(16:1/16:1)	BMP(14:0/14:0)	100.00
362	BMP(18:0_20:4)	BMP(14:0/14:0)	100.00
363	BMP(18:0_22:6)	BMP(14:0/14:0)	100.00
364	BMP(20:4/20:4)	BMP(14:0/14:0)	100.00
365	Cholesterol sulfate	(3-O-sulfo)GalCer(d18:1/18:0(d3))	0.47
366	CL(72:6/18:2)	CL(14:0/14:0/14:0/14:0)	100.00
367	CL(72:7/18:2)	CL(14:0/14:0/14:0/14:0)	100.00
368	CL(72:8-2(OOH)/18:2)	CL(14:0/14:0/14:0/14:0)	100.00
369	CL(72:8/18:2)	CL(14:0/14:0/14:0/14:0)	100.00
370	CL(74:9/18:2)	CL(14:0/14:0/14:0/14:0)	100.00
371	DHA	Arachidonic acid-d8_MRM	0.47
372	EPA	Arachidonic acid-d8_MRM	100.00
373	FAHFA(18:1/9-O-18:0)	PG(15:0/18:1(d7))	100.00
374	GD1a/b(d36:1)	GM3(d18:1/18:0(d5))	0.16
375	GD3(d34:1)	GM3(d18:1/18:0(d5))	100.00

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

376	GD3(d36:1)	GM3(d18:1/18:0(d5))	100.00
377	GM3(d34:1)	GM3(d18:1/18:0(d5))	100.00
378	GM3(d36:1)	GM3(d18:1/18:0(d5))	100.00
379	GQ1b(d36:1)	GM3(d18:1/18:0(d5))	100.00
380	Hemi-BMP(18:1/18:1)_16:0	Hemi-BMP(14:0/14:0)_14:0	100.00
381	Hemi-BMP(18:1/18:1)_18:0	Hemi-BMP(14:0/14:0)_14:0	100.00
382	Hemi-BMP(18:1/18:1)_18:1	Hemi-BMP(14:0/14:0)_14:0	100.00
383	Hemi-BMP(20:4/20:4)_16:0	Hemi-BMP(14:0/14:0)_14:0	100.00
384	Hemi-BMP(20:4/20:4)_18:0	Hemi-BMP(14:0/14:0)_14:0	100.00
385	Hemi-BMP(20:4/20:4)_18:1	Hemi-BMP(14:0/14:0)_14:0	100.00
386	Hemi-BMP(20:4/20:4)_20:4	Hemi-BMP(14:0/14:0)_14:0	100.00
387	Hemi-BMP(22:6/22:6)_16:0	Hemi-BMP(14:0/14:0)_14:0	100.00
388	Hemi-BMP(22:6/22:6)_18:0	Hemi-BMP(14:0/14:0)_14:0	100.00
389	Hemi-BMP(22:6/22:6)_18:1	Hemi-BMP(14:0/14:0)_14:0	100.00
390	Hemi-BMP(22:6/22:6)_22:6	Hemi-BMP(14:0/14:0)_14:0	100.00
391	Linoleic acid	Arachidonic acid-d8	100.00
392	Linolenic acid	Arachidonic acid-d8	100.00
393	LPE(16:0)	LPE(18:1(d7))	0.00
394	LPE(18:0)	LPE(18:1(d7))	0.16
395	LPG(16:0)	LPE(18:1(d7))	0.16
396	LPG(18:0)	LPE(18:1(d7))	38.12
397	LPG(18:1)	LPE(18:1(d7))	0.16
398	LPG(20:4)	LPE(18:1(d7))	100.00
399	LPG(22:6)	LPE(18:1(d7))	100.00
400	LPI(16:0)	LPE(18:1(d7))	0.31
401	LPI(18:0)	LPE(18:1(d7))	0.00
402	Oleic acid	Arachidonic acid-d8	0.31
403	PA(16:0_18:1)	PA(15:0/18:1(d7))	0.00
404	PA(18:0_18:1)	PA(15:0/18:1(d7))	100.00
405	PA(18:0_20:4)	PA(15:0/18:1(d7))	0.00
406	PA(18:0_22:6)	PA(15:0/18:1(d7))	0.00

PARKINSON'S PROGRESSION MARKERS INITIATIVE

Play a Part in Parkinson's Research

407	PA(18:1/18:1)	PA(15:0/18:1(d7))	100.00
408	Palmitic acid	Arachidonic acid-d8	0.31
409	Palmitoleic acid	Arachidonic acid-d8	100.00
410	PE(O-16:0/20:4)	PE(15:0/18:1(d7))	0.16
411	PE(O-16:0/22:6)	PE(15:0/18:1(d7))	0.31
412	PE(O-18:0/20:4)	PE(15:0/18:1(d7))	0.00
413	PE(O-18:0/22:6)	PE(15:0/18:1(d7))	0.00
414	PE(P-16:0/20:4)	PE(15:0/18:1(d7))	0.16
415	PE(P-16:0/20:5)	PE(15:0/18:1(d7))	100.00
416	PE(P-16:0/22:4)	PE(15:0/18:1(d7))	0.16
417	PE(P-16:0/22:6)	PE(15:0/18:1(d7))	0.16
418	PE(P-18:0/18:1)	PE(15:0/18:1(d7))	0.16
419	PE(P-18:0/18:2)	PE(15:0/18:1(d7))	0.00
420	PE(P-18:0/20:4)	PE(15:0/18:1(d7))	0.16
421	PE(P-18:0/20:5)	PE(15:0/18:1(d7))	65.47
422	PE(P-18:0/22:6)	PE(15:0/18:1(d7))	0.16
423	PE(P-18:1/20:4)	PE(15:0/18:1(d7))	0.16
424	PE(P-18:1/22:6)	PE(15:0/18:1(d7))	0.16
425	PEth(16:0_18:1)	PE(15:0/18:1(d7))	100.00
426	PEth(18:1/18:1)	PE(15:0/18:1(d7))	100.00
427	PG(16:0_18:1)	PG(15:0/18:1(d7))	2.34
428	PG(16:0_20:4)	PG(15:0/18:1(d7))	100.00
429	PG(16:0_22:6)	PG(15:0/18:1(d7))	100.00
430	PG(18:0_18:1)	PG(15:0/18:1(d7))	1.72
431	PG(18:0_20:4)	PG(15:0/18:1(d7))	100.00
432	PG(18:0_22:6)	PG(15:0/18:1(d7))	100.00
433	PG(18:1/18:1)	PG(15:0/18:1(d7))	100.00
434	PI(16:0_18:1)	PI(15:0/18:1(d7))	0.31
435	PI(16:0_20:4)	PI(15:0/18:1(d7))	0.16
436	PI(16:0_22:6)	PI(15:0/18:1(d7))	62.66
437	PI(18:0_18:1)	PI(15:0/18:1(d7))	0.78

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

438	PI(18:0_20:4)	PI(15:0/18:1(d7))	0.16
439	PI(18:0_22:6)	PI(15:0/18:1(d7))	0.16
440	PI(18:1/18:1)	PI(15:0/18:1(d7))	0.78
441	PI(20:4/20:4)	PI(15:0/18:1(d7))	100.00
442	PS(16:0_18:1)	PS(15:0/18:1(d7))	100.00
443	PS(16:0_20:4)	PS(15:0/18:1(d7))	100.00
444	PS(16:0_22:6)	PS(15:0/18:1(d7))	100.00
445	PS(18:0_18:1)	PS(15:0/18:1(d7))	0.00
446	PS(18:0_20:4)	PS(15:0/18:1(d7))	0.16
447	PS(18:0_22:6)	PS(15:0/18:1(d7))	0.16
448	PS(18:1/18:1)	PS(15:0/18:1(d7))	1.56
449	PS(22:6/22:6)	PS(15:0/18:1(d7))	100.00
450	Stearic acid	Arachidonic acid-d8	100.00

6. ASSAY: HALO_POSITIVE

Row	TESTNAME	IS_NAME	% missing
451	GlcCer(d18:1/16:0)	GlcCer(d18:1(d5)/18:0)	0.00
452	GlcCer(d18:1/18:0)	GlcCer(d18:1(d5)/18:0)	0.00
453	GlcCer(d18:2/18:0)	GlcCer(d18:1(d5)/18:0)	100.00
454	GlcCer(d18:1/20:0)	GlcCer(d18:1(d5)/18:0)	100.00
455	GlcCer(d18:2/20:0)	GlcCer(d18:1(d5)/18:0)	100.00
456	GlcCer(d18:1/22:0)	GlcCer(d18:1(d5)/18:0)	0.31
457	GlcCer(d18:1/22:1)	GlcCer(d18:1(d5)/18:0)	100.00
458	GlcCer(d18:2/22:0)	GlcCer(d18:1(d5)/18:0)	100.00
459	GlcCer(d18:1/24:1)	GlcCer(d18:1(d5)/18:0)	100.00
460	GlcCer(d18:1/24:0)	GlcCer(d18:1(d5)/18:0)	100.00
461	Glucosylsphingosine	Glucosylsphingosine(d5)	100.00
462	Cholesteryl glucoside	GlcCer(d18:1(d5)/18:0)	100.00
463	Cholesteryl galactoside	GlcCer(d18:1(d5)/18:0)	100.00
464	Sitosteryl glucoside	GlcCer(d18:1(d5)/18:0)	100.00
465	Sitosteryl galactoside	GlcCer(d18:1(d5)/18:0)	100.00

PARKINSON'S PROGRESSION MARKERS INITIATIVE
Play a Part in Parkinson's Research

466	GalCer(d18:1/16:0)	GalCer(d18:1/15:0)	0.00
467	GalCer(d18:1/18:0)	GalCer(d18:1/15:0)	0.00
468	GalCer(d18:2/18:0)	GalCer(d18:1/15:0)	0.16
469	GalCer(d18:1/20:0)	GalCer(d18:1/15:0)	0.16
470	GalCer(d18:2/20:0)	GalCer(d18:1/15:0)	0.16
471	GalCer(d18:1/22:0)	GalCer(d18:1/15:0)	0.16
472	GalCer(d18:1/22:1)	GalCer(d18:1/15:0)	0.00
473	GalCer(d18:2/22:0)	GalCer(d18:1/15:0)	0.16
474	GalCer(d18:1/24:1)	GalCer(d18:1/15:0)	0.00
475	GalCer(d18:1/24:0)	GalCer(d18:1/15:0)	0.00
476	Galactosylsphingosine	Glucosylsphingosine(d5)	100.00
477	alpha-GalCer(d18:1/16:0)	GalCer(d18:1/15:0)	0.16
478	alpha-GalCer(d18:1/18:0)	GalCer(d18:1/15:0)	100.00
479	alpha-GalCer(d18:2/:18:0)	GalCer(d18:1/15:0)	100.00
480	alpha-GalCer(d18:1/20:0)	GalCer(d18:1/15:0)	100.00
481	alpha-GalCer(d18:2/20:0)	GalCer(d18:1/15:0)	100.00
482	alpha-GalCer(d18:1/22:0)	GalCer(d18:1/15:0)	100.00
483	alpha-GalCer(d18:1/22:1)	GalCer(d18:1/15:0)	0.00
484	alpha-GalCer(d18:2/22:0)	GalCer(d18:1/15:0)	100.00
485	alpha-GalCer(d18:1/24:1)	GalCer(d18:1/15:0)	100.00
486	alpha-GalCer(d18:1/24:0)	GalCer(d18:1/15:0)	100.00

7. ASSAY: BMP_NEGATIVE

Row	TESTNAME	IS_NAME	% missing
487	BMP(22:6/22:6)	BMP(14:0/14:0)	0.47
488	BMP(18:1/18:1)	BMP(14:0/14:0)	0.31
489	BMP(18:1_22:6)	BMP(14:0/14:0)	22.19
490	BMP(18:0_18:1)	BMP(14:0/14:0)	0.16

8. ASSAY: GM_NEGATIVE

Row	TESTNAME	IS_NAME	% missing

491	GM3(d34:1)	GM3(d18:1/18:0(d5))	0.16
492	GM3(d36:1)	GM3(d18:1/18:0(d5))	0.31
493	GM3(d38:1)	GM3(d18:1/18:0(d5))	2.03
494	GM3(d40:1)	GM3(d18:1/18:0(d5))	1.25
495	GM2(d34:1)	GM3(d18:1/18:0(d5))	100.00
496	GM2(d36:1)	GM3(d18:1/18:0(d5))	87.50
497	GM2(d38:1)	GM3(d18:1/18:0(d5))	100.00
498	GM2(d40:1)	GM3(d18:1/18:0(d5))	100.00
499	GM1(d34:1)	GM3(d18:1/18:0(d5))	100.00
500	GM1(d36:1)	GM3(d18:1/18:0(d5))	87.50
501	GM1(d38:1)	GM3(d18:1/18:0(d5))	100.00
502	GM1(d40:1)	GM3(d18:1/18:0(d5))	100.00

Data Availability

Analyte measurements from all study samples and QC samples have been provided.

About the Authors

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